

AI Automation and Labor Market Outcomes[§]

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Abstract

Existing measures of occupational AI exposure ignore worker reallocation and fail to capture true welfare impacts. In this study, I address this by developing a general equilibrium framework that explicitly models the allocation problem for workers. Using administrative German data, I recover the unobserved and heterogeneous productivity vectors and switching cost structure that drive these choices. I then simulate an AI shock derived from task-level automation scores that leads to the new reallocation. The results show that the ability to reallocate determines the distribution of welfare gains. *Generalists*, defined by uniform comparative advantages, capture the largest relative wage gains by pivoting to high growth sectors. On the other hand, *specialists*, defined by concentrated comparative advantages, experience significantly smaller gains. This relative penalty affects both low skill workers trapped in certain tasks and high skill professionals forced into costly transitions, as the limited transferability of their skills restricts them from accessing higher returns in the labor market.

JEL Codes: J01, J20, J24, J62, E24, O33.

[§]The data access was provided via on-site use at the Research Data Centre (FDZ) of the German Federal Employment Agency (BA) at the Institute for Employment Research (IAB) and subsequently remote data access.

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1 Introduction

Automation technologies have had significant impact on the wage structure in the United States (Acemoglu and Restrepo (2022)). AI technologies that can potentially automate many tasks may have significant implications for the labor market (Trammell and Korinek (2023), McElheran et al. (2024)), especially considering the recent improvements in these technologies and the widening use cases (Bick, Blandin, and Deming (2024) and Handa et al. (2025)). There are several studies so far that measure AI exposure scores for occupations (among them, Brynjolfsson, Mitchell, and Rock (2018), Webb (2020), Felten, Raj, and Seamans (2018), Eloundou et al. (2023), and Handa et al. (2025)). However, the information content of these measures may be lacking in terms of welfare impacts, since they do not account for the worker reallocation resulting from the AI technologies shifting prices and labor demand. Rather than simply determining who is displaced, these general equilibrium effects shape how workers sort into new roles in response to changing incentives.

With this concern in mind, I study the impact of AI, specifically LLM technologies, on the wage distribution across workers and the employment distribution across occupations. I build a framework where workers can reallocate across occupations as a result of changing prices. Central to this analysis is the AI shock, which I measure using task-level automation scores following Eloundou et al. (2023) and Eisfeldt et al. (2023). I model this as a structural change that distorts relative occupational prices, which prompts workers to re-optimize their career choices based on their *comparative advantages* in the new environment.

Consistent with the recent automation literature (Acemoglu and Restrepo (2022), Acemoglu and Restrepo (2018), Acemoglu, Autor, et al. (2022), and Humlum (2021)), I model production at the task level. Each occupation consists of a set of tasks, some of which can be automated by AI technologies. I model automation as freeing human labor from automated tasks and reallocating it toward the remaining non-automated tasks within the occupation. This acts as an occupation-specific productivity shock. When a larger share of tasks is automated, the effective productivity of human labor in that occupation increases. On the supply side, workers sort across occupations in a dynamic discrete choice framework. In every period, forward-looking workers decide whether to remain in their current occupation or switch to a new one, subject to switching costs. Wages depend on observable characteristics and an unobserved productivity component, which I interpret as comparative advantage. Because comparative advantage is heterogeneous across both workers and occupations, two workers with identical observable traits may make different occupational choices.

Quantifying the welfare effects of this sorting requires recovering the unobserved comparative advantage vectors. Ideally, identification would rely on observing a worker’s wage history across all occupations to assess their potential productivity in every field. However, because individual employment histories are sparse, such comprehensive data is unavailable. To overcome this identification challenge, I model unobserved heterogeneity using finite latent worker types. Each type represents a group of workers who share a specific, unobserved

comparative advantage vector, which allows for the identification of the entire comparative advantage matrix. I estimate the type assignments and the structural wage parameters using an Expectation-Maximization (EM) algorithm following Arcidiacono and Miller (2011). To identify the switching costs separately from the wage parameters, I leverage the finite dependence property, which allows me to express the dynamic relationship between transition probabilities and value functions without solving the full dynamic programming problem at every step of the estimation.

For estimation, I use an administrative German panel data which tracks the employment history of workers between years 1998-2021¹. This panel dataset contains the employment status, occupational choice and wages of the 2% randomly selected sample of all individuals in Germany. The data includes the occupations and earnings history of workers. The large number of observed occupation-to-occupation transitions in this data allows me to identify the comparative advantage and switching cost parameters.

I find that the efficiency of reallocation is the key determinant of welfare in the new equilibrium. *Generalist* workers, those without a sharp comparative advantage in specific fields, capture the largest relative wage gains. Their reallocation ability allows them to fully capitalize on the productivity boom. In contrast, *specialist* workers, who possess a distinct comparative advantage in only one or a few occupations, experience smaller relative gains. These workers incur a high opportunity cost when switching, as they must forfeit the specific edge that defines their current productivity. This relative penalty affects both low-skill workers, whose comparative advantage is confined to manual or service tasks, and high-skill professionals forced to abandon specific domains, as the limited transferability of their skills restricts them from accessing the highest returns in the labor market.

I contribute to the automation literature, mainly to those concerning AI automation, including works such as Brynjolfsson, Mitchell, and Rock (2018), Eloundou et al. (2023), Felten, Raj, and Seamans (2018), Webb (2020), Humlum and Vestergaard (2024) and Handa et al. (2025), by studying the effects of the AI automation in a general equilibrium that incorporates worker reallocation. While the exposure scores are informative, there is no guaranteed one-to-one relationship between initial AI exposure and the welfare outcomes of workers in that occupation.

This study also adds to the previous works that study the general equilibrium effects of automation shocks using a reduced form analysis, such as Acemoglu and Restrepo (2022). In Acemoglu and Restrepo (2022), the authors estimate a propagation matrix that measures how an automation shock to a set of tasks ripples through to other tasks based on historical data. My contribution to this strand of literature is that I structurally estimate this propagation matrix, which depends on the primitives of the model environment, worker characteristics, and comparative advantage vectors. A reduced form approach is not feasible

¹While the raw data covers earlier years, I trim data to start from 1998. For a discussion on the data cleaning procedure, please see Section 4.

for the current AI shock because these technologies are in the early adoption phase, lacking sufficient historical data for estimation. A structural approach overcomes this limitation and provides the flexibility to quantify labor market responses to AI shock based on estimated structural parameters.

In parallel work, Smeets, Tian, and Traiberman (2025) study the AI shock using Danish data. Like this study, they account for worker reallocation using a dynamic discrete choice model to uncover comparative advantages. However, they find that lower-income workers are worse off in absolute terms, whereas I find that all groups experience wage gains, though to varying degrees. This divergence in distributional outcomes is likely driven by distinct modeling assumptions regarding the production structure. Specifically, Smeets, Tian, and Traiberman (2025) assume strong complementarity between sectors, following Atalay (2017). In a framework with strong complementarity, automation can lead to labor demand and wage contractions in specific sectors even under productivity growth. In contrast, I model AI as a strictly productivity-enhancing shock within a task-based framework, isolating the supply-side reallocation incentives in a setting where the marginal product of labor increases.

The rest of the paper is organized as follows. Section 2 presents the general equilibrium framework, detailing the task-based production hierarchy and the dynamic occupational choice problem. Section 3 describes the structural estimation of the wage switching cost parameters, and focuses on the identification of latent worker types. Section 4 outlines the administrative German panel data, the mapping procedure between German and US occupational codes, and the construction of the AI automation scores. Finally, Section 5 presents the estimation results and analyzes the counterfactual steady-state equilibrium under the AI productivity shock.

2 Theory

2.1 Production

The economy is populated by human workers normalized to unity. Time is discrete. Every period, each worker chooses among O different occupations to work. Each occupation consists of a series of tasks, where tasks can either be performed by human workers or the AI technology.

AI technology is a perfect substitute for human labor in the tasks it can perform, and it has zero rental cost. I model AI as infinitely scalable relative to human labor in automated tasks. Once a task is automated, human labor no longer needs to be allocated to that task. Instead, workers reallocate their time toward the remaining non-automated tasks within the occupation. This raises occupation-level output by increasing the effective amount of human labor applied to the tasks that still require workers, which is captured by the productivity multiplier M_o .

To illustrate the mechanism, consider the occupation of *translator*. Suppose AI performs the initial translation and the remaining human task is to review and edit the translation. In this case, occupational output depends on how efficiently human translators perform the remaining non-automated tasks. Automation raises effective productivity because the time previously spent on initial translation can be reallocated toward review, editing, and quality control.

The production technology for occupation o at time t is denoted by Y_{ot} and is assumed to have the following function form.

$$Y_{ot} = M_o \left[\sum_{\tau \in \tau_o \setminus \tau_A} Y_{\tau t}^{\frac{\theta-1}{\theta}} \right]^{\frac{\theta}{\theta-1}} \quad (1)$$

where τ_o denotes the set of tasks associated with occupation o and τ_A denotes the set of tasks that are automated. $M_o > 1$ is a productivity multiplier due to time reallocation of human workers from the automated tasks to the task they are performing. Denoting the share of automated tasks in occupation o by m_o , the formula productivity multiplier is as follows.

$$M_o = \frac{1}{1 - m_o} \quad (2)$$

Production in task τ is solely determined by human workers' productivity.

$$Y_{\tau t} = \sum_{n \in L_{ot}} z_{not} \quad (3)$$

where L_{ot} is the set of workers in occupation o at time t and z_{not} denotes the productivity of worker n in occupation o at time t .

Finally, the consumption basket of the households is a CES combination of the consumption of the individual occupation outputs.

$$c_{nt} = \left(\sum_{o=1}^O \mu_o^{\frac{1}{\rho}} c_{ont}^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \quad (4)$$

Here, μ_o denotes the occupation demand shifter for the occupation o output. Denoting the time t aggregate price level as P_t and the occupation o price level as P_{ot} , and along with the market clearing conditions, optimal consumption dictates:

$$Y_{ot} = \mu_o Y_t \left(\frac{P_{ot}}{P_t} \right)^{-\rho} \quad (5)$$

Each occupation is a perfectly competitive market. Workers are paid their marginal rev-

enue product, which equals the occupation output price multiplied by the worker's effective marginal product in that occupation.

$$w_{nt} = P_{o_{nt}} M_{o_{nt}} z_{no_{nt}t}. \quad (6)$$

Here, o_{nt} denotes worker n 's occupation at time t , $P_{o_{nt}}$ is the output price in that occupation, $M_{o_{nt}}$ is the occupation-specific productivity multiplier generated by automation, and $z_{no_{nt}t}$ is worker n 's productivity in that occupation.

2.2 Labor Supply

Having described labor demand, I now turn to labor supply. In each period, workers choose whether to remain in their current occupation or switch to another occupation. These choices depend on current wages, expected future values, deterministic switching costs, and switching cost shocks. Workers do not save, and they maximize expected lifetime utility:

$$\mathbb{E}_0 \sum_{t=0}^T \beta^t u(c_{nt}) \quad (7)$$

The flow utility of worker n who is working in occupation o_{nt} at time t is as follows.

$$u(c_{nt}) = w_{nt} + S_n(\xi_{nt}, o_{nt} | o_{nt-1}, \omega_{nt}) \quad (8)$$

where $S_n(\cdot)$ denotes the utility component associated with switching from occupation o_{nt-1} to o_{nt} . I refer to this term as the switching cost, with more negative values corresponding to larger costs. ω_{nt} denotes the worker characteristics that the switching costs depend on. For ease of expression, switching cost function can be separated into two terms as follows:

$$S_n(\xi_{nt}, o_{nt} | o_{nt-1}, \omega_{nt}) = s_n(o_{nt} | o_{nt-1}, \omega_{nt}) + \xi_{o_{nt}o_{nt-1}nt} \quad (9)$$

so that the deterministic part $s_n(o_{nt} | o_{nt-1}, \omega_{nt})$ and the stochastic part $\xi_{o_{nt}o_{nt-1}nt}$ are separated. For convenience, replace o_{nt} with o' and o_{nt-1} with o . Under certain conditions, worker n 's problem can be written as a Bellman equation as follows:

$$V_t(o, h_t, H_t) = \max_{o'} \{ \mathbb{E}_t w_{no't} + s_n(o' | o, \omega_{nt}) + \xi_{o'o_{nt}nt} + \beta \mathbb{E}_t V_{t+1}(o', h_{t+1}, H_{t+1}) \} \quad (10)$$

where h_t is the set of individual state and H_t is the set of aggregate state variables. $v_t(\cdot)$ is the value function associated with choosing a specific occupation at time t whereas $V_{t+1}(\cdot)$ is the value function associated with optimal occupation choice from time $t+1$ and on. There is an expectation term on w , since I assume the wage shocks are unobserved before making the occupation choice. Switching cost shocks, on the other hand, are observed before the occupation choice. I also assume that the switching cost shocks follow Type I extreme value

distribution ($\xi \sim F(0, \gamma)$) with scale parameter equal to γ and location parameter equal to 0. This yields the following recursive formulation (Rust (1987)).

$$v_t(o', h_t, H_t, \omega_{nt}) = \mathbb{E}_t w_{no't} + s_n(o'|o, \omega_{nt}) + \xi_{o'ont} + \gamma \int_{\xi} \log \sum_j \exp \left(\frac{\beta}{\gamma} v_{t+1}(j, h_{t+1}, H_{t+1}, \xi) \right) dF(\xi) + \beta \gamma c^e \quad (11)$$

where c^e is the Euler–Mascheroni constant. Time $t+1$ unconditional value function ($V_{t+1}(\cdot)$) can be manipulated to define a relationship between the value functions and the transition probabilities.

$$\begin{aligned} \mathbb{E}_t V_{t+1}(o', h_{t+1}, H_{t+1}, \omega_{nt+1}) &= \mathbb{E}_t [v_{t+1}(o', h_{t+1}, H_{t+1}, \omega_{nt+1}, \xi) \\ &\quad - \gamma \log \pi(o'|o', h_{t+1}, H_{t+1}, \omega_{nt+1})] + \gamma c^e \end{aligned} \quad (12)$$

This identity links value functions to conditional choice probabilities and is used below to recover switching cost parameters from observed transition probabilities and expected wage differentials.

2.3 State Variables

h_t represents the individual state variables, age, schooling category and the unobserved productivity parameters for the worker. In practice, estimating productivity parameters for each worker is not feasible because workers' employment histories are sparse. Instead, I assume that each worker belongs to one of the finite types $i \in \{1, \dots, I\}$. This assumption reduces the support problem because productivity profiles are estimated at the type level rather than separately for each worker. Identification then relies on having enough type-weighted observations across occupations. Due to the estimation procedure described in Section 3, all workers have strictly positive posterior probability of belonging to each type, which helps avoid exact zero-support cells. However, effective sample size can still be small for some type-occupation pairs, so the comparative advantage parameters are regularized through the lower-dimensional occupation skill representation introduced below.

H_t denotes the aggregate state variables. In this setting, the relevant aggregate states are occupational output prices. Workers can perfectly forecast the individual state variables, age and the time-invariant type, since they are non-stochastic. Imposing a forecasting method for the aggregate state variables (occupational prices) is not necessary for the estimation purposes, since I am following Arcidiacono and Miller (2011) which allows me to work with empirical transition rates.

2.4 Latent Productivity and Comparative Advantage

During the estimation period, where there is no AI automation yet, $M_o = 1$ for all occupations. The wage of worker n is therefore equal to the worker's productivity in the chosen occupation multiplied by the price of that occupation's output:

$$w_{nt} = P_{ont} z_{not}$$

where z_{not} is the productivity of worker n in occupation o at time t . Productivity is, in logarithmic form, a linear function of individual state variables.

$$\log z_{not} = \beta_1 \times \text{Age}_{nt} + \beta_2 \text{Age}_{nt}^2 + \beta_3 \text{Schooling}_n + a_{i(n)o_{nt}} + \sigma_o \varepsilon_{nt} \quad (13)$$

where a_{io} denotes the type-occupation absolute advantage component of log productivity for worker type i in occupation o . These absolute advantage components are not separately identified from occupation prices. The reason is that wages depend on the sum of log prices and log productivity. For any occupation o , adding a constant to all type-specific absolute advantage components a_{io} and subtracting the same constant from the log occupation price leaves all predicted wages unchanged. Thus, the data can identify only productivity differences across types within an occupation, not the absolute productivity level of every type in that occupation. I therefore normalize type 1 to be the benchmark type and define comparative advantage as

$$CA_{io} = a_{io} - a_{1o}, \quad CA_{1o} = 0$$

The estimated object is therefore the comparative advantage component CA_{io} , not the absolute advantage component a_{io} . The shock ε_{nt} is an idiosyncratic wage shock that is unobserved before the occupation choice and is independent of individual and aggregate state variables. Finally, σ_o denotes the occupation-specific standard deviation of wage shocks.

2.5 Equilibrium

This section lays out the equilibrium conditions for the post-AI economy. The pre-AI economy satisfies the same conditions, except that no tasks are automated, so $\tau_A = \emptyset$ and $M_o = 1$ for all occupations.

Workers choose occupations in a forward-looking manner. Their choices depend on current wages, deterministic switching costs, switching cost shocks, and expected future values. Because switching is costly, workers may wait for a favorable switching cost shock before

moving to another occupation.

$$\max_{o_{nt}} \mathbb{E}_0 \sum_{t=0}^T u(c_{nt}) \quad (14)$$

$$\text{s. to } c_{nt} = w_{nt} + S_n(\xi_{nt}, o_{nt} | o_{nt-1}, \omega_{nt}) \quad (15)$$

Let $V_t^n(o, h_t, H_t)$ denote worker n 's expected discounted lifetime utility when choosing occupation o at time t . The optimal occupation satisfies

$$o_{nt} \in \arg \max_{o \in \{1, \dots, O\}} \mathbb{E}_t V_t^n(o, h_t, H_t). \quad (16)$$

On the production side, firms combine task outputs into occupational output. Each occupation is perfectly competitive, so workers are paid their marginal revenue product. The zero-profit condition implies

$$P_{ot} = \frac{1}{M_o} \frac{w_{nt}}{z_{not}} \quad \forall n \in L_{ot}. \quad (17)$$

Since workers may differ in productivity within the same occupation, wages can differ across workers employed in the same occupation.

Labor market clearing condition is

$$\sum_{o=1}^O L_{ot} = 1, \quad \forall t. \quad (18)$$

Goods market clearing condition is

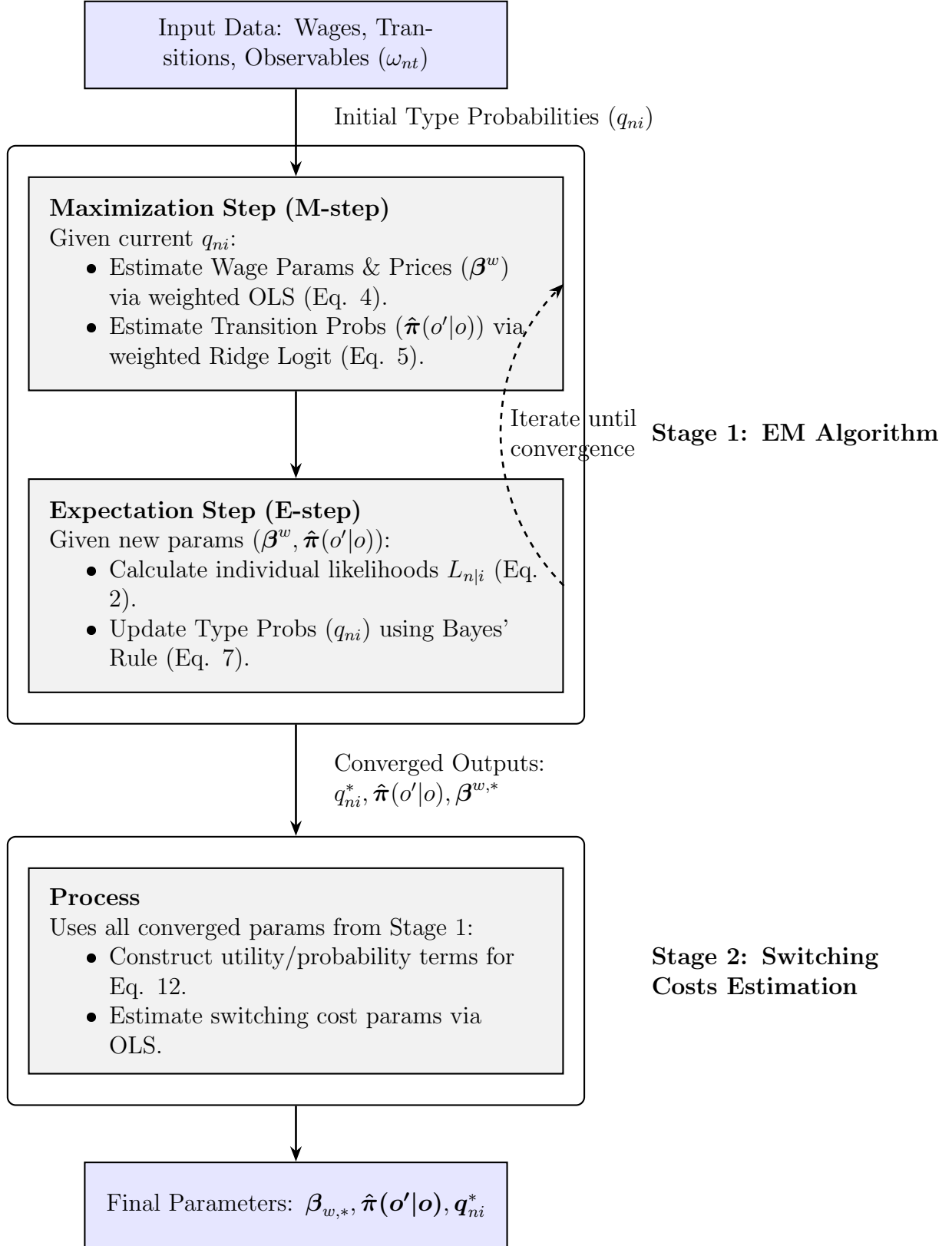
$$C_{ot} = Y_{ot}, \quad \forall o, t \quad (19)$$

3 Estimation

The estimation problem has three unknown objects: wage parameters, transition probabilities, and worker type probabilities. If worker types were observed, the wage equation and transition equations could be estimated conditional on type. Since worker types are unobserved, direct maximization would require jointly searching over model parameters and the type probability vector for every worker. I use the EM algorithm to break this problem into two simpler steps. Given current type probabilities, the maximization step estimates type-specific wage and transition parameters. Given these parameters, the expectation step updates each worker's posterior type probabilities using the likelihood of the worker's observed wage and occupation history.

The posterior probabilities have a direct interpretation. Workers receive higher probability

Figure 1: Visualized Estimation Procedure



of belonging to a type when their observed wages and transitions are more likely under that type's estimated parameters. For example, if one type has high comparative advantage in occupation 1 and low comparative advantage in occupation 2, then a worker who earns more than observationally similar workers in occupation 1 and less in occupation 2 receives a higher posterior probability of belonging to that type.

Switching costs are estimated after the first stage. The first stage recovers wage parameters, occupation prices, transition probabilities, and posterior type probabilities. The second stage then uses these estimates to recover switching cost parameters, as described in Section 3.2.

First, define the likelihood contribution for worker n in period t , conditional on type i :

$$L_{nt|i} = f(w_{nt} \mid o_{nt}, \omega_{nt}, i) \times \pi(o_{nt} \mid o_{n,t-1}, \omega_{nt}, i), \quad (20)$$

where $f(\cdot)$ is the conditional wage density and $\pi(\cdot)$ is the transition probability from last period's occupation to the current occupation. The vector ω_{nt} contains worker observables, age and schooling. The full likelihood contribution of worker n , conditional on type i , is

$$L_{n|i} = \prod_{t=1}^T f(w_{nt} \mid o_{nt}, \omega_{nt}, i) \times \pi(o_{nt} \mid o_{n,t-1}, \omega_{nt}, i). \quad (21)$$

The unconditional likelihood is the weighted sum of the type-specific likelihood contributions:

$$L = \prod_{n=1}^N \sum_{i=1}^I q_i(\omega_{n1}^{obs}) L_{n|i}, \quad (22)$$

where q_{ni} denotes the probability attached to worker n being type i .

A remaining complication is that $\pi(\cdot)$ depends on objects that are not directly observed. In the structural model, transition probabilities depend on comparative advantage, switching costs, and choice shocks. Therefore, maximizing the likelihood while recomputing structural transition probabilities would require solving the worker's dynamic problem repeatedly inside the EM algorithm. This is computationally infeasible.

To avoid this nested fixed point problem, I follow Arcidiacono and Miller (2011) and estimate the conditional choice probabilities directly from the observed transition data. The first stage therefore treats transition probabilities as empirical objects rather than recomputing them from the dynamic model at every EM iteration. The next subsection describes this first stage, and Figure 1 summarizes the full estimation procedure.

3.1 First Stage

The first stage starts from an initial posterior type probability vector for each worker. I use occupation histories to initialize these probabilities. Occupations are grouped into four broad categories, and each worker’s initial type probabilities are based on the share of the worker’s observed history spent in each category. I impose a lower bound on each type probability so that the initialization is informative but not deterministic. This breaks the symmetry across types while still allowing the likelihood to update type probabilities in later EM iterations.

3.1.1 Maximization Step

The maximization step estimates the wage equation by weighted least squares and the transition probabilities by weighted L2-regularized logit regressions. Occupation prices are estimated as part of the wage equation. Since occupation prices are common across worker types, I estimate the wage equation for all types jointly, imposing a single occupation-year price for each occupation and period.

With 34 occupations and 4 types, estimating unrestricted comparative advantage would require 34×3 type-occupation parameters, since type 1 is the normalization group. To reduce dimensionality, I parameterize comparative advantage using occupation characteristics. I use quantified occupation characteristics from O*NET, such as the importance of mathematics, reading comprehension, and negotiation, and summarize them using the first eight principal components. These eight components explain 82.5% of the variance in the O*NET information matrix.

The comparative advantage component is parameterized as

$$CA_{io} = \mathbb{1}\{i \neq 1\} \Gamma_o \beta_{CA}^i, \quad (23)$$

where Γ_o is the vector of occupation characteristics and β_{CA}^i is the type-specific skill vector. Substituting this expression into the log wage equation gives

$$\log w_{nt} = p_{ont} + \beta_1 \text{Age}_{nt} + \beta_2 \text{Age}_{nt}^2 + \beta_3 \text{Schooling}_n + \mathbb{1}\{i \neq 1\} \Gamma_{ont} \beta_{CA}^i + \sigma_o \varepsilon_{nt}. \quad (24)$$

This equation is estimated by weighted least squares after stacking the data by type, with posterior type probabilities q_{ni} entering as observation weights.

For transition probabilities, the conditional choice probability approach suggests estimating the empirical distribution of transitions directly. To this end, I use a bin estimator as follows:

$$\pi_i(h_2, o_2 \mid h_1, o_1) = \frac{\sum_{n=1}^N \sum_{t=2}^T q_{ni} \mathbb{1}\{h_{nt} = h_2, o_{nt} = o_2, h_{n,t-1} = h_1, o_{n,t-1} = o_1\}}{\sum_{n=1}^N \sum_{t=2}^T q_{ni} \mathbb{1}\{h_{n,t-1} = h_1, o_{n,t-1} = o_1\}}. \quad (25)$$

Where h denotes the individual state variables. In practice, this bin estimator is not a reliable measure because partitioning the data based on individual state variables and occupation result with very few observations for some (age, schooling and occupation) triplets. Traiberman (2019) faces the same problem and uses a linear probability model to approximate the bin estimator while Ransom (2022) uses a logit estimator. I find that both approaches lead to numerical instabilities in my case. With an LPM model, numerical instabilities arise because the predicted probabilities are not bound between 0 and 1. With the logit model, some estimations do not converge given very few observations for some transitions. Therefore, I rely on an L2-regularized (ridge) logit to keep the parameter estimates from taking implausible values when there are only a few observations for a transition.

3.1.2 Expectation Step

In the expectation step, posterior type probabilities are updated using Bayes' rule. If a worker's observed history has a higher likelihood under the parameters of type i , then the posterior probability attached to type i increases. Formally,

$$q_{ni}^{(m+1)} = \frac{L_{n|i}^{(m)} q_i^{(m)}(\omega_{n1}^{obs})}{\sum_{i'=1}^I L_{n|i'}^{(m)} q_{i'}^{(m)}(\omega_{n1}^{obs})}. \quad (26)$$

The prior probability $q_i(\omega_{n1}^{obs})$ allows initial observed characteristics, age and schooling, to predict latent type membership, following Arcidiacono and Miller (2011).

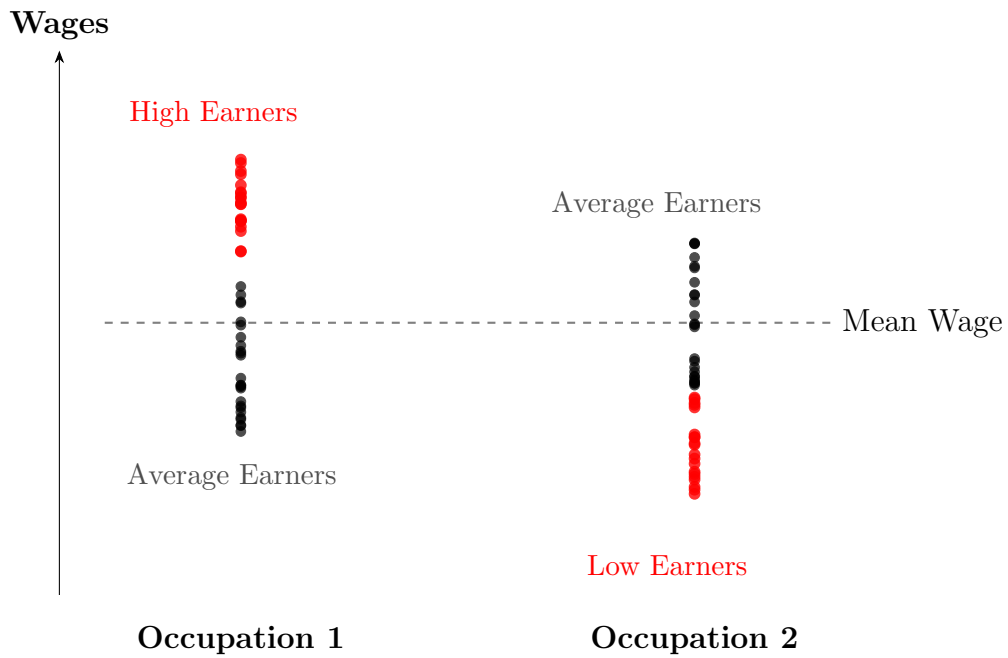
3.1.3 Identification of Worker Types

Identification of worker types relies on wages and occupational transitions. Workers who earn more than observationally similar workers in a given occupation are likely to have positive comparative advantage in that occupation. If the same workers display similar wage patterns across other occupations, these wage histories help separate latent types. Figure 2 illustrates this logic with two occupations.

Transition histories provide additional information because workers with different latent profiles may move through occupations differently. For example, two workers may spend similar amounts of time in broad occupation groups but differ in how they move between them. One worker may move directly between high- and low-skill occupations, while another may move gradually through intermediate occupations. These transition patterns help distinguish types even when occupation shares alone are similar. Figure 3 illustrates this source of identification.

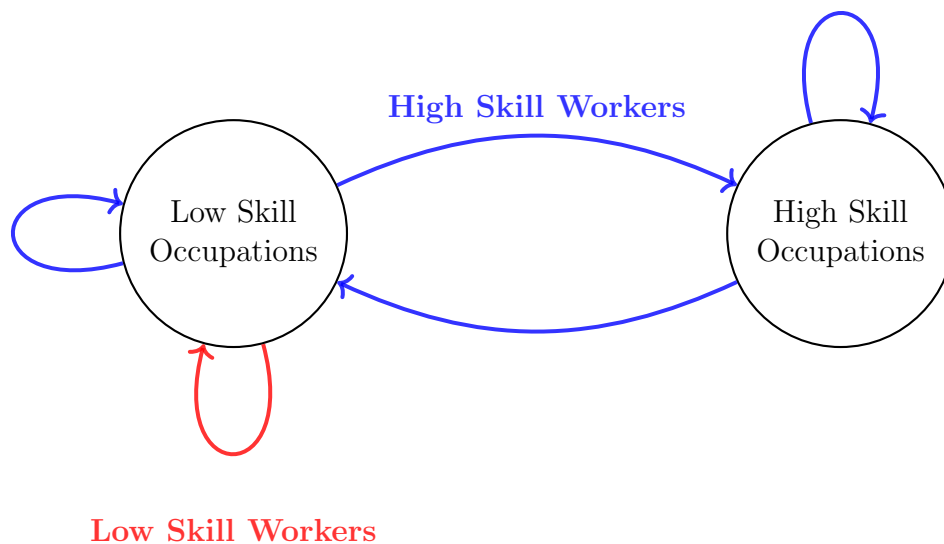
Identification of occupation prices relies on the same mobility patterns that help identify comparative advantage. If a type of worker is observed only in one occupation, the wage equation cannot separately distinguish that type's comparative advantage in the occupation from the occupation price. Price and comparative advantage estimates are therefore more

Figure 2: Worker Types and Comparative Advantage (CA).



Note: A group of workers (red dots) who earn more than average in Occupation 1 also earn less than average in Occupation 2, suggesting they belong to a specific unobserved *type*.

Figure 3: Worker Types and Transitions



Note: High skill workers (blue) can transition between both high and low skill occupations. Low skill workers (red) are restricted to transitions within low skill occupations. The transition histories are likely to separate this two groups of workers into different types.

robust when workers, especially workers with different latent profiles, are observed moving across occupations.

3.2 Second Stage

The second stage estimates switching cost parameters using the conditional choice probabilities recovered in the first stage. The identifying equation compares two transition odds that share the same destination occupation. Because the destination is the same, workers with the same state variables have the same continuation value after the transition. This allows the continuation value to be differenced out and expresses transition odds in terms of expected wage differences and switching costs. Section 1.A.2.3 provides the full derivation.

$$\begin{aligned} \log \left(\frac{\pi_t(o'|o, h_t, H_t, \omega_{nt})}{\pi_t(o|o, h_t, H_t, \omega_{nt})} \right) + \beta \log \left(\frac{\pi_{t+1}(o'|o', h_{t+1}, H_{t+1}, \omega_{nt+1})}{\pi_{t+1}(o|o, h_{t+1}, H_{t+1}, \omega_{nt+1})} \right) \\ = \frac{1}{\gamma} \mathbb{E}_t [w_{no't} - w_{not}] + \frac{1 - \beta}{\gamma} s_n(o'|o, \omega_{nt}) + \frac{1}{\gamma} [\xi_{o'ont} - \xi_{o'ont+1}] \end{aligned} \quad (27)$$

Transition probabilities and expected wage differentials are recovered in the first stage. Following the first stage, I estimate this equation via OLS where the discount factor $\beta = 0.96$.

Estimating a separate switching cost for every occupation pair would require many observed transitions for each pair. Since some occupation pairs have few transitions, I follow Ransom (2022) and Traiberman (2019) and parameterize switching costs as a function of distance between occupations:

$$s_n(o'|o, \omega_{nt}) = \alpha_0^o + \alpha_1 d(o, o'). \quad (28)$$

Here, α_0^o is an origin-occupation fixed cost and $d(o, o')$ measures the distance between occupations o and o' . I construct this distance using the Euclidean distance between the principal-component vectors associated with each occupation. Since the principal components are ordered by explained variance, this distance gives more weight to components with greater variation across occupations.

4 Data

4.1 Labor Panel

I use German administrative data from the Sample of Integrated Labour Market Biographies, which contains a 2 percent random sample from all individuals in Germany (Graf et al. (2023)). The data covers the period between 1975 and 2021, include worker identifiers,

age, employment status, occupation, schooling, gender, and income. For schooling, I define two categories, university education and all the other lower degrees because the data do not provide a finer measure of schooling.

I use observations from 1998 onward.² The occupation classification is KldB 2010. While observations after November 2011 are reported directly using KldB 2010, earlier observations use KldB 1998 and are converted to KldB 2010 by the data provider.

I use O*NET to measure occupation characteristics. O*NET reports task descriptions and attributes related to skills, knowledge, work activities, and abilities. O*NET uses the SOC 2019 classification, and there is no direct crosswalk between SOC 2019 and KldB 2010. I therefore map SOC 2019 occupations to ISCO-08 and then map ISCO-08 occupations to KldB 2010.

To construct the principal components, I follow Traiberman (2019) and use the O*NET categories “Knowledge,” “Skills,” “Work Activities,” and “Abilities.” For each attribute, O*NET reports “Importance” and “Level.” Following Firpo, Fortin, and Lemieux (2011), I combine these two measures using geometric weights of $2/3$ and $1/3$, respectively. This leaves 160 attributes for each occupation. I then apply principal component analysis and retain the first eight components. Appendix Section 1.A.3 reports the most important attributes for these components.

4.2 AI Exposure Scores

To compute the post-AI equilibrium, I need a measure of which tasks can be automated by AI in each occupation. I follow Eloundou et al. (2023) and Eisefeldt et al. (2023) and classify O*NET tasks using an LLM. For each task description, I ask the model whether the task can be automated by current LLM-based technologies. This produces a binary automation score at the task level.

I aggregate task-level scores to occupations by taking the average automation score across tasks within each O*NET occupation. I then use the SOC–ISCO–KldB crosswalk described above to aggregate these scores to 2-digit KldB 2010 occupations. Table 1.A.1.2 reports the resulting occupation-level AI exposure scores.³

²After 1998, employers started reporting earnings below a threshold. I also restrict the sample period for computational reasons. The estimation requires stacking the wage regression matrix by type in order to estimate type-specific comparative advantage parameters while imposing common occupation prices. This substantially increases memory requirements, so I further restrict the estimation sample by randomly selecting worker IDs after the data cleaning process.

³I use a system prompt to define the model’s role in classifying task automation. The prompt follows Eisefeldt et al. (2023), with modifications reflecting recent advances in LLM capabilities. The exact prompt is reported in Appendix Section 1.A.4.

5 Results

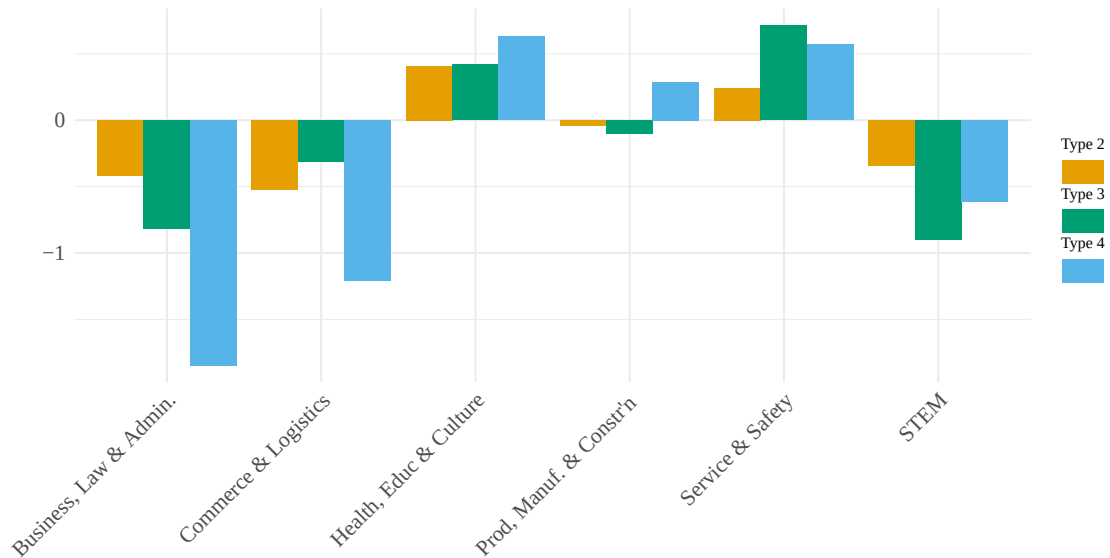
I first discuss the estimation results, i.e., Mincer coefficients, comparative advantage profiles, type distributions, and switching costs. These objects summarize the estimated worker heterogeneity and mobility frictions before the AI shock. I then turn to the post-AI equilibrium, where occupational productivity changes affect wages, prices, and worker allocations.

Figure 4 shows the comparative advantage profiles of the three non-normalized types across occupation groups.⁴ Type 1 is the normalization group, so its comparative advantage is zero in every occupation. The figure should therefore be interpreted as showing whether each type is more or less productive than type 1 within each occupation group.

Relative to type 1, type 2 and type 3 have flatter comparative advantage profiles and therefore resemble all-rounder types. Type 4 has a more concentrated profile, with relative strength in Services & Safety, Health, Education & Culture, and Production, Manufacturing & Construction, and weaker relative productivity in the remaining occupation groups.

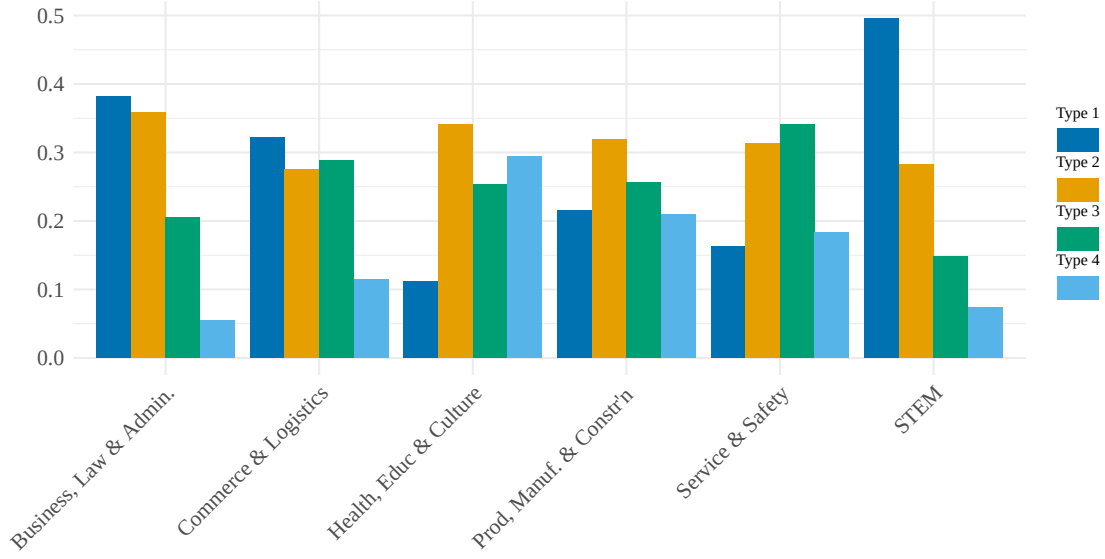
Figures 5 and 6 show how estimated types are distributed across occupation groups. The patterns are consistent with sorting based on comparative advantage. Type 4 workers account for a relatively large share of employment in the occupation groups where their comparative advantage is high, while type 2 and type 3 workers display less concentrated occupation distributions, consistent with their flatter comparative advantage profiles.

Figure 4: Comparative Advantage Across Occupation Groups



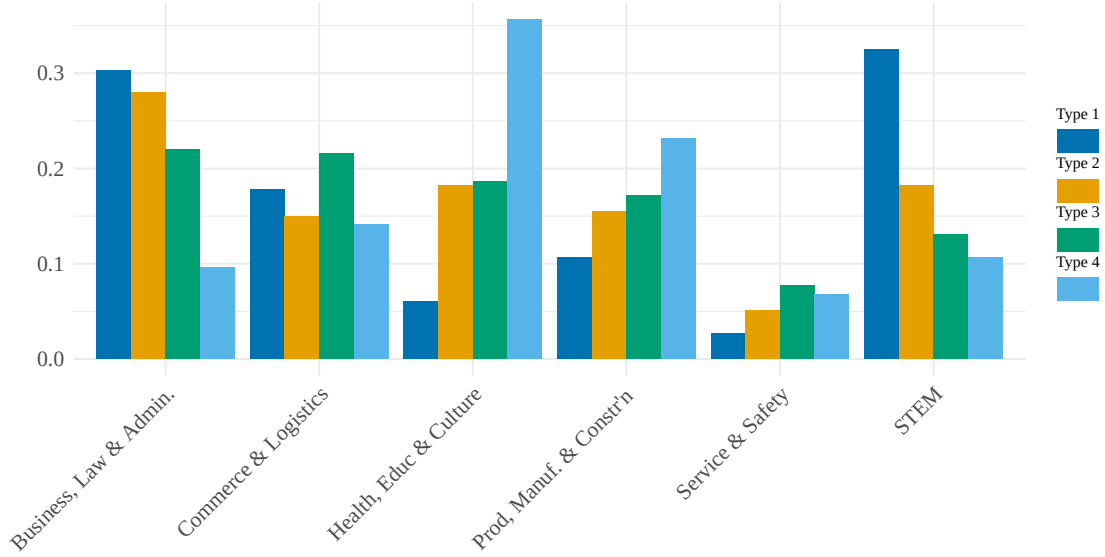
⁴For the list of occupations and the corresponding occupation groups, see Table 1.A.1.3.

Figure 5: Share of Types Across Occupation Groups



Note: Total shares of all types for each occupation group are 1.

Figure 6: Distribution of Types Across Occupations



Note: Occupation shares for each type sum to 1.

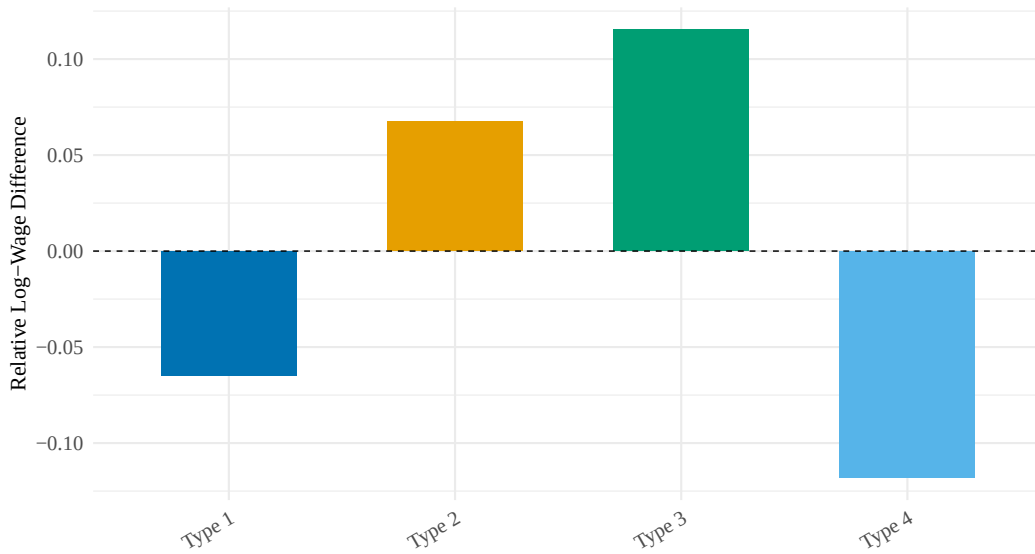
5.1 Post-AI Equilibrium

I compare the estimated pre-AI equilibrium with a counterfactual post-AI equilibrium. In both cases, I start from the average estimated occupation prices over the last three sample years. The only direct change is the occupation-specific productivity multiplier M_o , which increases in occupations with a larger share of automatable tasks. I then solve for the

new equilibrium prices and worker allocation using the algorithm described in Algorithm 1.A.2.2.

Because AI enters as a positive productivity shock, average wages increase. However, the gains are uneven across worker types.⁵ Figure 7 shows that type 1 and type 4 workers experience smaller relative gains than type 2 and type 3 workers. The flatter comparative advantage profiles of type 2 and type 3 workers make it less costly for them to move toward occupations where AI raises productivity more strongly.

Figure 7: Relative Log-Wage Differences Across Types



Notes: The y -axis reports the difference between post-AI and pre-AI log wages, relative to the unweighted average wage difference across types.

Figure 8 shows a positive relationship between occupational price changes and the share of tasks that remain non-automated in the post-AI equilibrium.⁶ Since AI raises productivity more in occupations with larger automated task shares, output expands more in those occupations, putting downward pressure on their prices. In the counterfactual equilibrium, the direct productivity effect on wages is larger than the offsetting price decline, so employment does not fall sharply in highly exposed occupations.

Figure 9 shows the change in worker allocation across occupation groups. The largest reallocation is from Manufacturing & Construction, a relatively low-exposure group, toward Business, Law & Administration, a relatively high-exposure group. This movement is driven mainly by type 1 workers.

This pattern reflects heterogeneity in comparative advantage. Workers move into expanding

⁵Type-level wage and employment statistics are computed using posterior type probabilities as weights. I also check robustness using hard type assignments based on posterior probability thresholds, and the qualitative patterns are unchanged.

⁶Although the average occupation price is higher in the post-AI equilibrium, the price of the final consumption basket is normalized to be the same across equilibria.

Figure 8: Changes in Occupation Prices vs. Non-Automated Task Shares



Notes: Left y -axis is for the occupational price differences between the post-AI equilibrium and the original equilibrium. Right y -axis is for the inverse automated task shares.

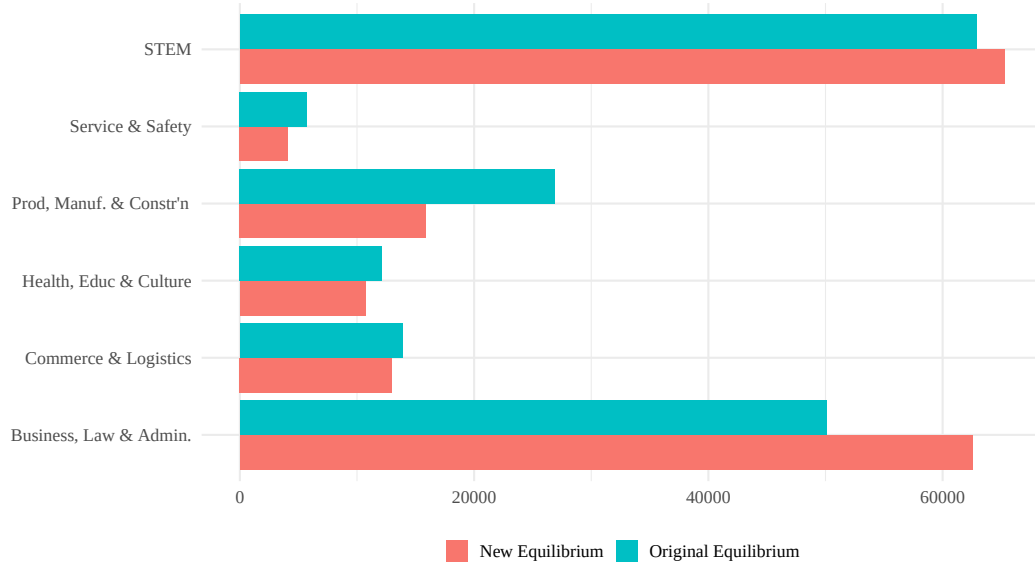
occupations when the productivity gain and option value of switching outweigh the loss in occupation specific comparative advantage. Type 1 workers account for much of the inflow because, relative to other estimated types, they are not strongly disadvantaged in the expanding occupation groups. However, they do not receive the largest relative wage gains because the destination occupations also experience price declines.

Figure 10 compares reallocation intensity and relative wage changes by type. Type 1 workers exhibit the largest change in their occupational distribution, but their wage gains are modest. This suggests that their mobility is partly a response to declining relative returns in their initial occupations rather than movement toward occupations where they have especially strong comparative advantage. Type 2 and type 3 workers experience larger relative wage gains with less reallocation, consistent with their flatter comparative advantage profiles. Type 4 workers show the least reallocation, reflecting more concentrated comparative advantage in occupations that are less affected by the AI shock.

6 Conclusion

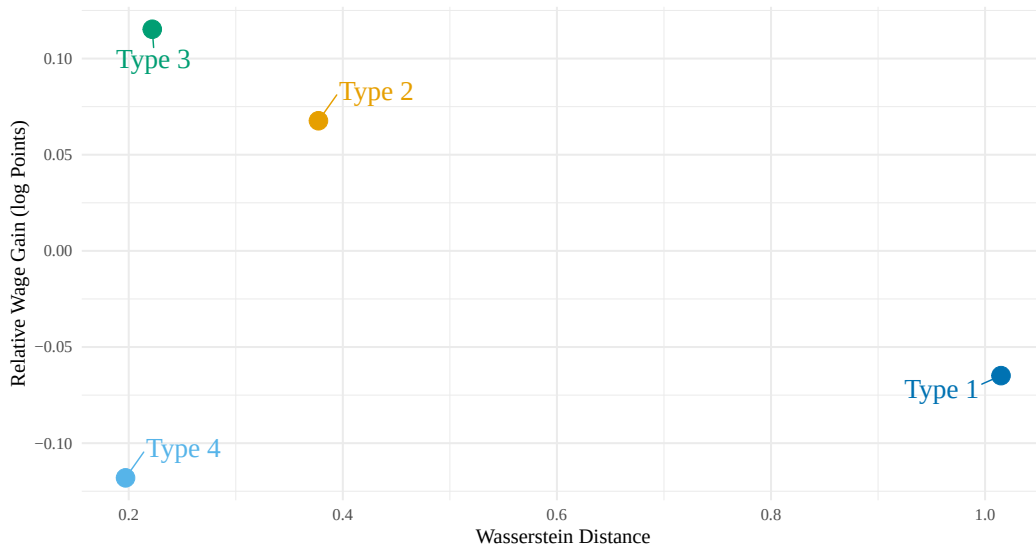
As AI technology becomes increasingly integrated into the workplace, the future of the labor market remains uncertain. To address this uncertainty, many predictions regarding occupational exposure to AI have been developed. However, these predictions offer an incomplete picture, as they fail to account for worker reallocation. Consequently, existing exposure scores are difficult to interpret in isolation. In this study, I build a framework where production can be performed by either human workers or AI technology. To analyze these

Figure 9: Worker Allocation Across the Occupation Groups



Notes: x -axis denote the number of workers.

Figure 10: Displacement vs. Relative Wage Change



Notes: x -axis denote the Wasserstein distance between the pre and post-AI type distributions across occupations.

reallocation dynamics, I estimate a model of occupational choice featuring heterogeneous worker types with varying comparative advantages across occupations.

I estimate four worker types that differ in how concentrated their comparative advantage profiles are across occupations. Some types have relatively flat profiles, while others have more concentrated relative strengths. I find that comparative advantage and type shares across occupations are positively related, suggesting sorting with respect to comparative advantage.

Using task-level automation scores constructed from O*NET task descriptions, I simulate an AI productivity shock and solve for the counterfactual post-AI equilibrium. The results show that AI raises average wages, but the gains differ across worker types. Workers with flatter comparative advantage profiles benefit more because they can move toward occupations with larger productivity gains at lower productivity cost. Workers with more concentrated comparative advantage profiles are less able to capture gains outside the occupations where they are relatively productive.

While AI automation will inevitably generate divergent economic outcomes, accurate welfare assessment requires accounting for the margin of worker reallocation. This study demonstrates that a worker's current occupation is an insufficient statistic for predicting their post-shock trajectory. Instead, the ultimate distribution of gains is governed by unobserved comparative advantages and the capacity to move across occupations. Consequently, as AI adoption accelerates, policy frameworks must look beyond static occupational labels and prioritize understanding the labor mobility.

Appendix to Chapter 1

1.A.1 Additional Tables & Figures

Table 1.A.1.1: Wage Equation Estimates by Worker Type

	All Types	Type 1	Type 2	Type 3	Type 4
Age	0.0419 (0.0001)				
Age ²	-0.0004 (0.0000)				
Schooling $\leq$ Secondary	-0.1822 (0.0002)				
1 st PC		0	-0.0499 (0.0003)	-0.2102 (0.0003)	-0.1739 (0.0004)
2 nd PC		0	-0.1046 (0.0003)	-0.0713 (0.0003)	0.0269 (0.0004)
3 rd PC		0	0.1575 (0.0004)	0.5647 (0.0004)	0.3885 (0.0005)
4 th PC		0	0.1794 (0.0008)	0.3952 (0.0008)	-0.0704 (0.0010)
5 th PC		0	-0.3357 (0.0008)	-0.6689 (0.0008)	-1.2483 (0.0012)
6 th PC		0	0.8912 (0.0009)	0.6521 (0.0009)	1.5685 (0.0011)
7 th PC		0	-0.9100 (0.0012)	-0.0996 (0.0014)	-1.8305 (0.0017)
8 th PC		0	-0.1419 (0.0011)	-1.3706 (0.0012)	-0.3703 (0.0015)

Notes: Standard errors are reported in parentheses. The “All Types” column shows coefficients common to all worker types. Principal components (PC) are type-specific transformations of occupation characteristics. Type 1 coefficients for principal components are normalized to zero. All coefficients are statistically significant at the 1% level.

Table 1.A.1.2: Exposure Scores by Occupation

Occupation	Score	Occupation	Score
Advertising, marketing, and media design	0.63	Tourism, hotels, and restaurants	0.34
Financial services, accounting, and tax	0.61	Non-medical healthcare and body care	0.34
Computer science and ICT	0.59	Metal production and construction	0.33
Business management and organization	0.59	Raw materials, glass, and ceramic processing	0.31
Humanities, social sciences, and economics	0.56	Medical and health care	0.31
Purchasing, sales, and trading	0.54	Technical building services	0.31
Law and public administration	0.49	Teaching and training	0.30
Construction planning and surveying	0.49	Cleaning services	0.29
Technical R&D, construction, and production	0.44	Mechatronics and electrical engineering	0.27
Safety, security, and surveillance	0.42	Plastic, wood, and wood processing	0.26
Math, biology, chemistry, and physics	0.42	Gardening and floristry	0.25
Traffic and logistics	0.41	Machine-building and automotive technology	0.23
Retail sales	0.40	Education, social work, and theology	0.21
Agriculture, forestry, and farming	0.36	Food production and processing	0.21
Paper, printing, and technical media design	0.35	Vehicle and transport equipment operation	0.21

Note: Scores indicate the share of tasks within the occupation that are predicted to be automatable.

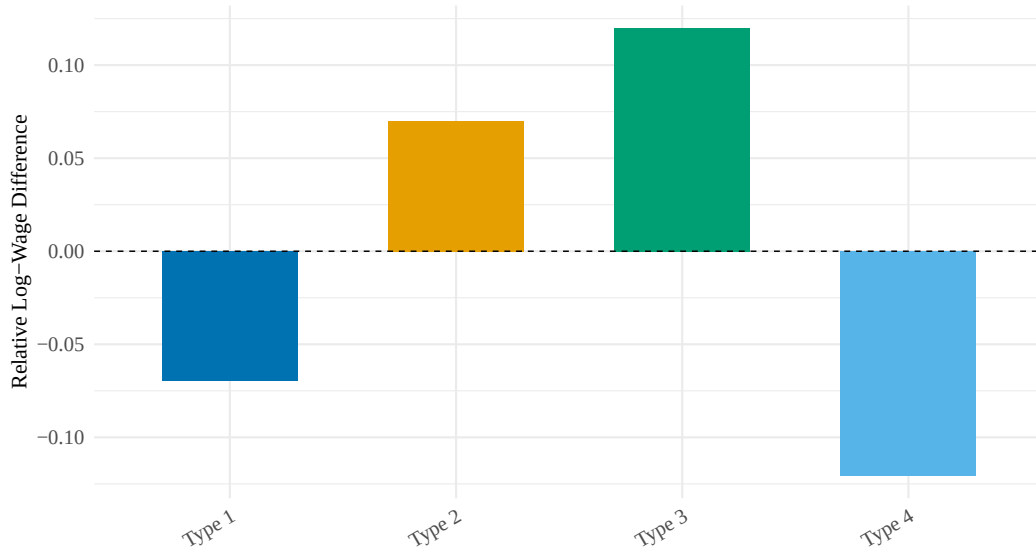
Table 1.A.1.3: Occupation Groups

Occupation	Group
Business management and organization	Business, Law & Admin.
Financial services, accounting, and tax	Business, Law & Admin.
Law and public administration	Business, Law & Admin.
Traffic and logistics	Commerce & Logistics
Vehicle and transport equipment operation	Commerce & Logistics
Purchasing, sales, and trading	Commerce & Logistics
Retail sales	Commerce & Logistics
Medical and health care	Health, Educ & Culture
Non-medical healthcare and body care	Health, Educ & Culture
Education, social work, and theology	Health, Educ & Culture
Teaching and training	Health, Educ & Culture
Humanities, social sciences, and economics	Health, Educ & Culture
Advertising, marketing, and media design	Health, Educ & Culture
Product design, craftwork, and fine arts	Health, Educ & Culture
Raw materials, glass, and ceramic processing	Prod, Manuf. & Construction
Plastic, wood, and wood processing	Prod, Manuf. & Construction
Paper, printing, and technical media design	Prod, Manuf. & Construction
Metal production and construction	Prod, Manuf. & Construction
Machine-building and automotive technology	Prod, Manuf. & Construction
Textile and leather production	Prod, Manuf. & Construction
Food production and processing	Prod, Manuf. & Construction
Building construction	Prod, Manuf. & Construction
Interior construction	Prod, Manuf. & Construction
Technical building services	Prod, Manuf. & Construction
Agriculture, forestry, and farming	Service & Safety
Gardening and floristry	Service & Safety
Safety, security, and surveillance	Service & Safety
Cleaning services	Service & Safety
Tourism, hotels, and restaurants	Service & Safety
Mechatronics and electrical engineering	STEM
Technical R&D, construction, and production	STEM
Construction planning and surveying	STEM
Math, biology, chemistry, and physics	STEM
Computer science and ICT	STEM

Table 1.A.1.4: Switching Cost / Average Wage by Occupation

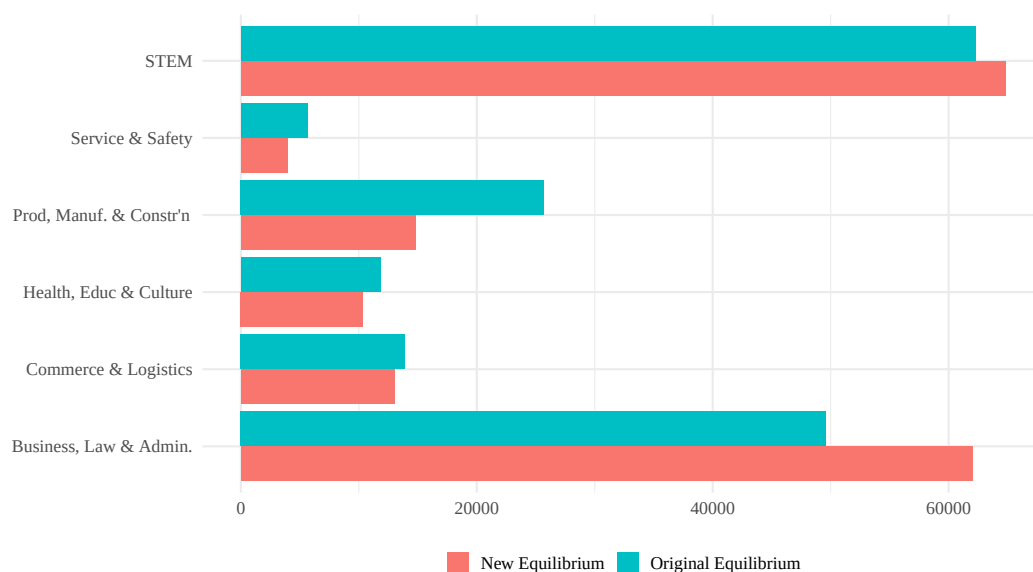
Occupation	$\bar{S}_n(\cdot o)/\bar{w}$	Occupation	$\bar{S}_n(\cdot o)/\bar{w}$
Agriculture, forestry, and farming	0.163	Traffic and logistics	0.524
Gardening and floristry	0.208	Vehicle and transport equipment operation	0.422
Raw materials, glass, and ceramic processing	0.178	Safety, security, and surveillance	0.352
Plastic, wood, and wood processing	0.380	Cleaning services	0.388
Paper, printing, and technical media design	0.264	Purchasing, sales, and trading	0.504
Metal production and construction	0.631	Retail sales	0.471
Machine-building and automotive technology	0.697	Tourism, hotels, and restaurants	0.334
Mechatronics and electrical engineering	0.506	Business management and organization	1.012
Technical R&D, construction, and production	0.559	Financial services, accounting, and tax	0.910
Textile and leather production	0.163	Vehicle and transport equipment operation	0.537
Food production and processing	0.408	Medical and health care	0.612
Construction planning and surveying	0.355	Non-medical healthcare and body care	0.297
Building construction	0.367	Education, social work, and theology	0.348
Interior construction	0.268	Teaching and training	0.262
Technical building services	0.436	Humanities, social sciences, and economics	0.262
Math, biology, chemistry, and physics	0.400	Advertising, marketing, and media design	0.480
Computer science and ICT	0.902	Product design, craftwork, and fine arts	0.169

Figure 1.A.1.1: Relative Log-Wages Across Types - Probability Threshold = 0.5



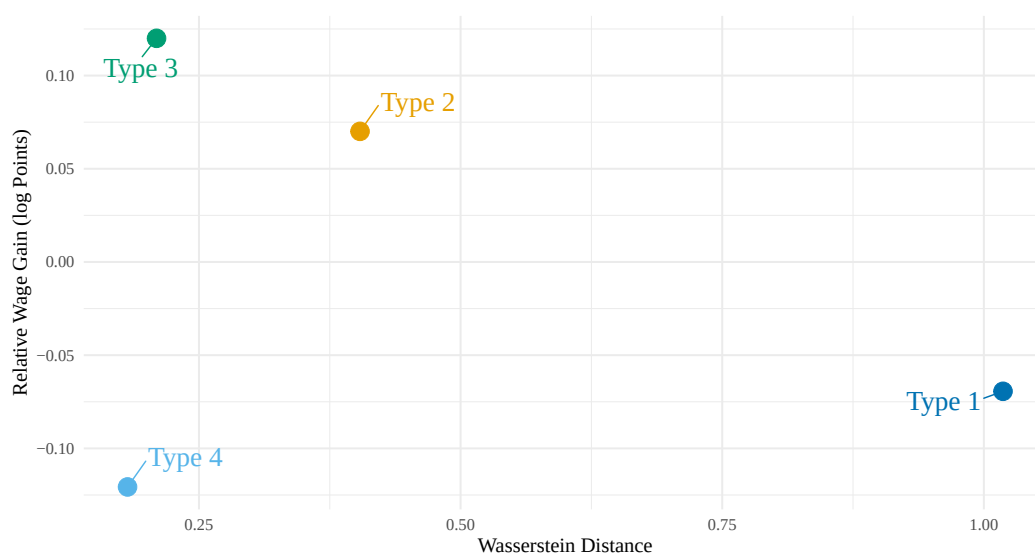
Notes: Posterior type probabilities meeting or exceeding the threshold are assigned a probability of 1, otherwise 0. y -axis indicate the difference between the wages in the post-AI and pre-AI equilibria, relative to the (unweighted) average wage difference across types. The picture with the average weighted by the type shares still look similar since type 1 and type 4 workers have a total of 45% share in the economy.

Figure 1.A.1.2: Worker Allocation Across the Occupation Groups - Probability Threshold = 0.5



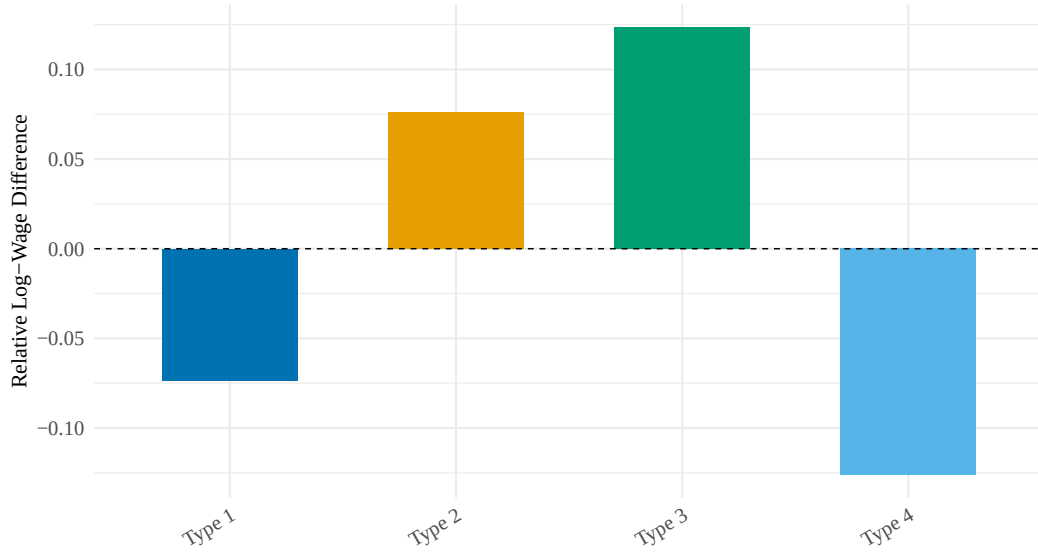
Notes: Posterior type probabilities meeting or exceeding the threshold are assigned a probability of 1, otherwise 0. x -axis denote the number of workers.

Figure 1.A.1.3: Displacement vs. Relative Wage Change - Probability Threshold = 0.5



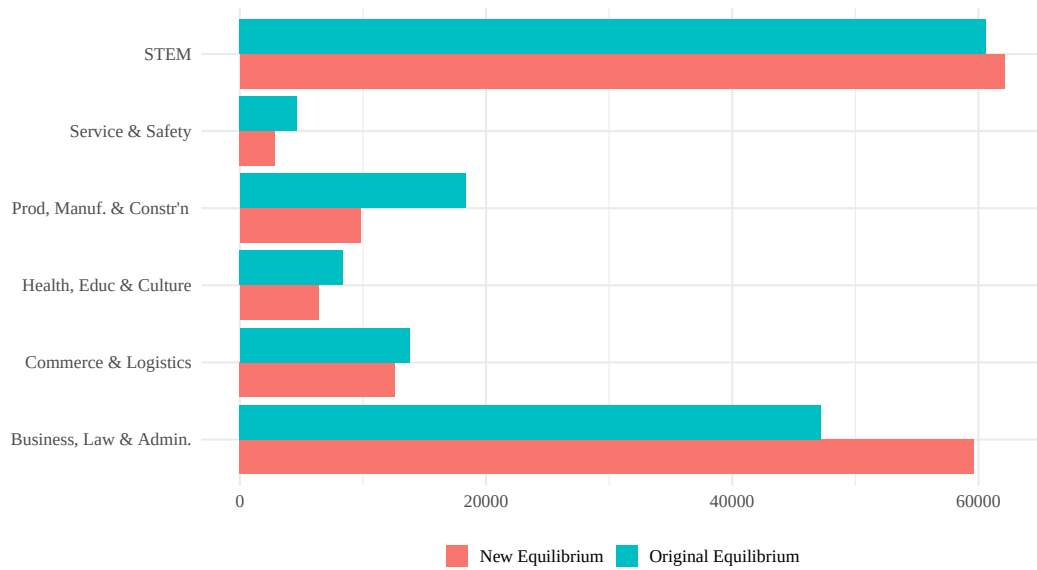
Notes: x -axis denote the Wasserstein distance between the pre and post-AI type distributions across occupations.

Figure 1.A.1.4: Relative Log-Wages Across Types - Probability Threshold = 0.7



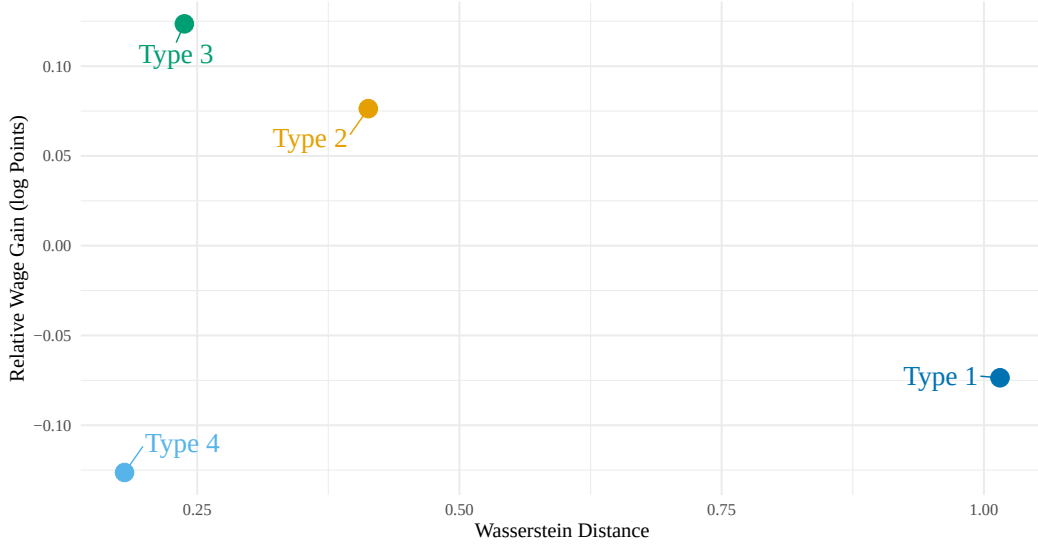
Notes: Posterior type probabilities meeting or exceeding the threshold are assigned a probability of 1, otherwise 0. y -axis indicate the difference between the wages in the post-AI and pre-AI equilibria, relative to the (unweighted) average wage difference across types. The picture with the average weighted by the type shares still look similar since type 1 and type 4 workers have a total of 45% share in the economy.

Figure 1.A.1.5: Worker Allocation Across the Occupation Groups - Probability Threshold = 0.7



Notes: Posterior type probabilities meeting or exceeding the threshold are assigned a probability of 1, otherwise 0. x -axis denote the number of workers.

Figure 1.A.1.6: Displacement vs. Relative Wage Change - Probability Threshold = 0.7



Notes: x -axis denote the Wasserstein distance between the pre and post-AI type distributions across occupations.

1.A.2 Estimation Appendix

1.A.2.1 EM Algorithm

The section follows Section 3 to provide more details on the estimation procedure. The exact algorithm for the EM estimation is provided in Algorithm 1.A.2.1.

EM estimator needs an initial type probability vector. I partition the occupations in 4 groups (the same as 1.A.1.3), and construct the probability vector based on the share of histories spent on each occupation groups. For example, a worker who spent all of their career in a single occupation group would be assigned an initial probability of 0.7 to the type initially associated with that group, and the 0.3 probability would be uniformly allocated to the other types.

I use $\lambda = 1e - 2$ for the L2-regularization. After the log-likelihood converges, I re-estimate the parameters for the transition probabilities, this time with no penalty ($\lambda = 0$).

$$\boldsymbol{\beta}^{i,\pi,*} := \arg \max_{\boldsymbol{\beta}^{i,\pi}} \mathbf{q}_i^{*'} [\mathbf{d} \log \mathbf{p}^{i,\pi} + (\mathbf{1} - \mathbf{d}) \log(\mathbf{1} - \mathbf{p}^{i,\pi})], \quad \mathbf{p}^{i,\pi} = \Lambda(X\boldsymbol{\beta}^{i,\pi}). \quad (1.A.2.6)$$

Then estimate the predicted transition probabilities for the second stage.

$$\hat{\pi}(o'|o) := \frac{\exp(X\boldsymbol{\beta}_{o'}^{i,\pi,*})}{1 + \sum_j \exp(X\boldsymbol{\beta}_j^{i,\pi,*})} \quad (1.A.2.7)$$

Algorithm 1.A.2.1 Expectation-Maximization (EM) Algorithm for the First Stage

- 1: Initialize type probabilities $q_{ni}^{(0)}$ for all n, i .
- 2: **while** the total log-likelihood has not converged **do**
Maximization Step
- 3: Estimate transition parameters $\beta^{i,\pi,*}$ for each type i via weighted L2-regularized logit:

$$\beta^{i,\pi,*} := \arg \max_{\beta^{i,\pi}} \left\{ \mathbf{q}_i^{(m)'} [\mathbf{d} \log \mathbf{p}^{i,\pi} + (\mathbf{1} - \mathbf{d}) \log(\mathbf{1} - \mathbf{p}^{i,\pi})] - \frac{\lambda}{2} \|\beta^{i,\pi}\|^2 \right\}, \quad (1.A.2.1)$$

where

$$\mathbf{p}^{i,\pi} = \Lambda(X\beta^{i,\pi})$$

and $\Lambda(\cdot)$ is the logistic CDF.

- 4: Estimate wage parameters $\beta^{i,w,*}$ for each type i via weighted OLS:

$$\beta^{i,w,*} := \arg \min_{\beta^{i,w}} \left\{ (\mathbf{w} - X\beta^{i,w})' Q_i^{(m)} (\mathbf{w} - X\beta^{i,w}) \right\}, \quad (1.A.2.2)$$

where $Q_i^{(m)} = \text{diag}(\mathbf{q}_i^{(m)})$.

- 5: Estimate type probability regression parameters $\beta^{q,*}$ via weighted multinomial logit:

$$\beta^{q,*} := \arg \max_{\beta^q} \sum_{n=1}^N \sum_{i=1}^I q_{ni}^{(m)} \log q(i|\omega_{n1}^{obs}; \beta^q). \quad (1.A.2.3)$$

Expectation Step

- 6: Calculate individual likelihoods $L_{n|i}$ using the new parameters $(\beta^{i,\pi,*}, \beta^{i,w,*})$.
- 7: Generate predicted type probabilities:

$$\hat{q}_{ni}^{prior} = q(i|\omega_{n1}^{obs}; \beta^{q,*}). \quad (1.A.2.4)$$

- 8: Update worker-specific type probabilities using Bayes' rule:

$$q_{ni}^{(m+1)} = \frac{L_{n|i} \hat{q}_{ni}^{prior}}{\sum_{i'=1}^I L_{n|i'} \hat{q}_{ni'}^{prior}}. \quad (1.A.2.5)$$

- 9: **end while**
-

Expected wage differentials can also be recovered from the wage regression.

$$\widehat{\mathbb{E}_t w_{no't}} - \widehat{\mathbb{E}_t w_{not}} = X_{nt} \boldsymbol{\beta}_{no'}^{i,w} - X_{nt} \boldsymbol{\beta}_{no}^{i,w}. \quad (1.A.2.8)$$

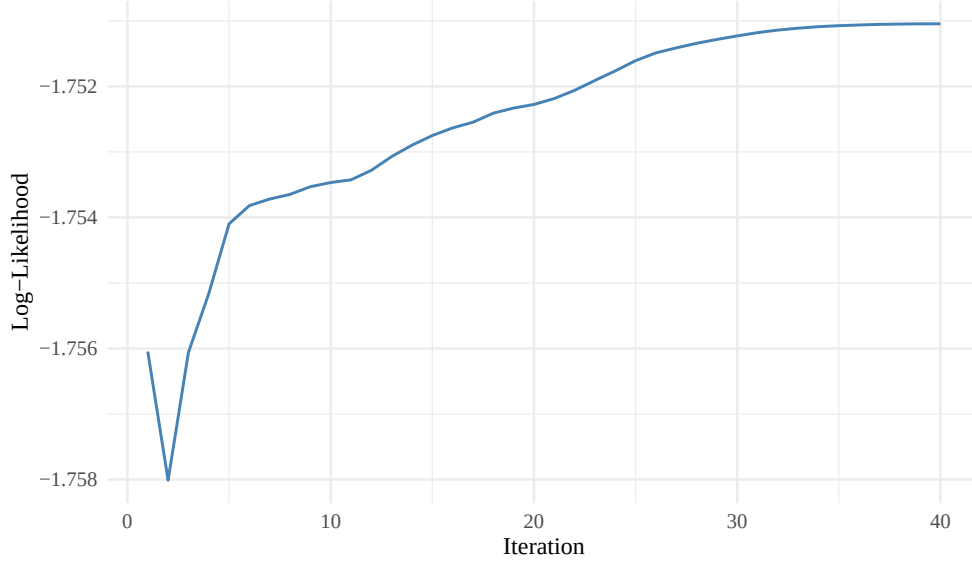
Given Eq. 1.A.2.7 and Eq. 1.A.2.8 and the distance measure between the occupations, I have all the variables for the second stage regression. Hence, I calculate the parameters for the scale parameter (γ) for the switching cost shocks and the distance-cost multiplier (α_1) as well as the fixed cost for switching from each occupation ($\{\alpha_o^0\}_{o=1}^O$).

1.A.2.2 Log-Likelihood

Figure 1.A.2.1 reports the first-stage log-likelihood across EM iterations. I stop the EM algorithm when the improvement in the log-likelihood falls below $1e - 6$.

The log-likelihood is nearly monotone over the EM iterations. The only visible decline occurs immediately after the first update of posterior type probabilities. This early decline is likely due to numerical instability in the transition-probability step, where weighted L2-regularized logits are estimated using type probabilities that are still far from their converged values. After this initial adjustment, the log-likelihood increases steadily until convergence.

Figure 1.A.2.1: Log-Likelihood for the First Stage



Note: At 40th iteration, the improvement in the log-likelihood is less than $1e - 6$.

1.A.2.3 Derivation of the Regression Formula for the Second Stage

This section derives the regression equation used to estimate switching cost parameters in the second stage. The derivation uses the logit structure implied by the Type I extreme value

switching cost shocks. The key step is to express differences in continuation values using conditional choice probabilities, so that switching costs can be estimated from observed transition probabilities and expected wage differences.

Let $v_t(o'|o, h_t, H_t, \omega_{nt})$ denote the deterministic component of the value of choosing occupation o' at time t , given previous occupation o and state variables (h_t, H_t, ω_{nt}) . This object excludes the Type I extreme value switching cost shock. The realized value of choosing o' is therefore

$$v_t(o'|o, h_t, H_t, \omega_{nt}) + \xi_{o'ont}. \quad (1.A.2.9)$$

The deterministic choice specific value is

$$v_t(o'|o, h_t, H_t, \omega_{nt}) = \mathbb{E}_t w_{no't} + s_n(o'|o, \omega_{nt}) + \beta \mathbb{E}_t V_{t+1}(o', h_{t+1}, H_{t+1}, \omega_{nt+1}). \quad (1.A.2.10)$$

Because the switching cost shocks are Type I extreme value, the conditional choice probabilities satisfy

$$\pi_t(o'|o, h_t, H_t, \omega_{nt}) = \frac{\exp(v_t(o'|o, h_t, H_t, \omega_{nt})/\gamma)}{\sum_j \exp(v_t(j|o, h_t, H_t, \omega_{nt})/\gamma)}. \quad (1.A.2.11)$$

Therefore, for any two choices o' and o ,

$$\gamma \log \left(\frac{\pi_t(o'|o, h_t, H_t, \omega_{nt})}{\pi_t(o|o, h_t, H_t, \omega_{nt})} \right) = v_t(o'|o, h_t, H_t, \omega_{nt}) - v_t(o|o, h_t, H_t, \omega_{nt}). \quad (1.A.2.12)$$

Substituting the deterministic choice-specific values gives

$$\begin{aligned} \gamma \log \left(\frac{\pi_t(o'|o, h_t, H_t, \omega_{nt})}{\pi_t(o|o, h_t, H_t, \omega_{nt})} \right) &= \mathbb{E}_t [w_{no't} - w_{not}] + s_n(o'|o, \omega_{nt}) - s_n(o|o, \omega_{nt}) \\ &\quad + \beta [\mathbb{E}_t V_{t+1}(o', h_{t+1}, H_{t+1}, \omega_{nt+1}) - \mathbb{E}_t V_{t+1}(o, h_{t+1}, H_{t+1}, \omega_{nt+1})]. \end{aligned} \quad (1.A.2.13)$$

I normalize the deterministic cost of staying in the same occupation to zero, so $s_n(o|o, \omega_{nt}) = 0$. The remaining term is the continuation value difference. To express it in terms of transition probabilities, use the logit inversion at time $t + 1$, taking o' as the reference choice. When the previous occupation at $t + 1$ is o' , the value of choosing o' satisfies

$$\begin{aligned} \mathbb{E}_t V_{t+1}(o', h_{t+1}, H_{t+1}, \omega_{nt+1}) &= \mathbb{E}_t [w_{no',t+1} + \beta V_{t+2}(o', h_{t+2}, H_{t+2}, \omega_{nt+2})] \\ &\quad - \gamma \log \pi_{t+1}(o'|o', h_{t+1}, H_{t+1}, \omega_{nt+1}) + \gamma c^e. \end{aligned} \quad (1.A.2.14)$$

When the previous occupation at $t + 1$ is o , the value of choosing o' satisfies

$$\begin{aligned}\mathbb{E}_t V_{t+1}(o, h_{t+1}, H_{t+1}, \omega_{nt+1}) &= \mathbb{E}_t [w_{no',t+1} + s_n(o'|o, \omega_{nt+1}) + \beta V_{t+2}(o', h_{t+2}, H_{t+2}, \omega_{nt+2})] \\ &\quad - \gamma \log \pi_{t+1}(o'|o, h_{t+1}, H_{t+1}, \omega_{nt+1}) + \gamma c^e.\end{aligned}\quad (1.A.2.15)$$

Subtracting the second equation from the first cancels the future wage term, the continuation value, and the Euler constant term:

$$\begin{aligned}\mathbb{E}_t V_{t+1}(o', h_{t+1}, H_{t+1}, \omega_{nt+1}) - \mathbb{E}_t V_{t+1}(o, h_{t+1}, H_{t+1}, \omega_{nt+1}) \\ = -s_n(o'|o, \omega_{nt+1}) - \gamma \log \left(\frac{\pi_{t+1}(o'|o', h_{t+1}, H_{t+1}, \omega_{nt+1})}{\pi_{t+1}(o'|o, h_{t+1}, H_{t+1}, \omega_{nt+1})} \right).\end{aligned}\quad (1.A.2.16)$$

Substituting this continuation value difference into the time t log odds equation gives

$$\begin{aligned}\gamma \log \left(\frac{\pi_t(o'|o, h_t, H_t, \omega_{nt})}{\pi_t(o|o, h_t, H_t, \omega_{nt})} \right) &= \mathbb{E}_t [w_{no't} - w_{not}] + s_n(o'|o, \omega_{nt}) - \beta s_n(o'|o, \omega_{nt+1}) \\ &\quad - \beta \gamma \log \left(\frac{\pi_{t+1}(o'|o', h_{t+1}, H_{t+1}, \omega_{nt+1})}{\pi_{t+1}(o'|o, h_{t+1}, H_{t+1}, \omega_{nt+1})} \right).\end{aligned}\quad (1.A.2.17)$$

Since switching costs do not depend on time-varying worker characteristics, we have $s_n(o'|o, \omega_{nt}) = s_n(o'|o, \omega_{nt+1})$. The equation simplifies to

$$\begin{aligned}\gamma \log \left(\frac{\pi_t(o'|o, h_t, H_t, \omega_{nt})}{\pi_t(o|o, h_t, H_t, \omega_{nt})} \right) &= \mathbb{E}_t [w_{no't} - w_{not}] + (1 - \beta) s_n(o'|o, \omega_{nt}) \\ &\quad - \beta \gamma \log \left(\frac{\pi_{t+1}(o'|o', h_{t+1}, H_{t+1}, \omega_{nt+1})}{\pi_{t+1}(o'|o, h_{t+1}, H_{t+1}, \omega_{nt+1})} \right).\end{aligned}\quad (1.A.2.18)$$

Moving the future probability term to the LHS and dividing by γ gives the regression equation:

$$\begin{aligned}\log \left(\frac{\pi_t(o'|o, h_t, H_t, \omega_{nt})}{\pi_t(o|o, h_t, H_t, \omega_{nt})} \right) + \beta \log \left(\frac{\pi_{t+1}(o'|o', h_{t+1}, H_{t+1}, \omega_{nt+1})}{\pi_{t+1}(o'|o, h_{t+1}, H_{t+1}, \omega_{nt+1})} \right) \\ = \frac{1}{\gamma} \mathbb{E}_t [w_{no't} - w_{not}] + \frac{1 - \beta}{\gamma} s_n(o'|o, \omega_{nt}) + u_{not}.\end{aligned}\quad (1.A.2.19)$$

The error term u_{not} contains the realized difference in switching cost shocks that is not included in the deterministic choice specific value. It also captures approximation error from replacing model implied transition probabilities and expected wage differences with their first stage estimates. I estimate this equation by OLS, fixing $\beta = 0.96$.

1.A.2.4 Construction of the Post-AI Equilibrium

I first recover the preference shifters μ_o from expenditure shares. Starting from Equation 5, the occupation-level demand system implies

$$\mu_o = \frac{Y_{ot}}{Y_t} \left(\frac{P_{ot}}{P_t} \right)^\rho \quad (1.A.2.20)$$

Multiplying and dividing the right hand side by $(P_t/P_{ot})^{\rho-1}$, the consumption preference shifters can be expressed in terms of expenditure shares, individual occupation prices and the aggregate price level.

$$\mu_o = \frac{Y_{ot}}{Y_t} \frac{P_{ot}}{P_t} P_{ot}^{1-\rho} P_t^{\rho-1} \quad (1.A.2.21)$$

The first term is the expenditure share of occupation o . Since there is no direct data source for occupational output prices, I use the occupation prices and quantities recovered from the estimation. The aggregate price index P_t is not separately needed because the preference shifters are normalized to sum to one. I therefore recover μ_o up to scale and normalize $\sum_o \mu_o = 1$. I set the elasticity of substitution across occupational outputs to $\rho = 1.78$, following Burstein, Morales, and Vogel (2019).⁷

The iterative algorithm used to compute the counterfactual steady state is described in Algorithm 1.A.2.2. Given a vector of occupational prices, workers solve the dynamic occupation choice problem implied by the estimated wage and switching cost parameters. Because switching cost shocks are observed before occupation choice and follow a Type I extreme value distribution, the algorithm uses the corresponding log-sum value function and updates worker allocations using the implied choice probabilities. New allocations determine occupation outputs, which imply a new vector of occupational prices. I iterate on this process until occupational prices converge.

The price update uses damping to avoid oscillations in the fixed point iteration. Without damping, large changes in worker allocation can generate large price movements, which may reverse switching incentives in the next iteration. The algorithm therefore updates effective prices gradually and stops when the maximum absolute change in log occupational prices is below the convergence tolerance.

1.A.3 Data Appendix

1.A.3.1 Data Preparation

The raw data contain more than 77 million observations after episode splitting and more than 55 million observations before episode splitting. The data provider performs episode

⁷Burstein, Morales, and Vogel (2019) estimate this parameter using U.S. data and 30 occupations.

Algorithm 1.A.2.2 Equilibrium Solver for Occupational Prices and Worker Allocation

- 1: *Initialization*
- 2: Start from an initial worker allocation $o_n^{(0)}$.
- 3: Calculate occupation outputs $Y_o^{(0)}$, aggregate output $Y^{(0)}$, and initial occupational prices

$$P_{o,true}^{(0)} = \mu_o^{1/\rho} \left(\frac{Y_o^{(0)}}{Y^{(0)}} \right)^{-1/\rho}.$$

- 4: Set effective prices $P_o^{(0)} \leftarrow P_{o,true}^{(0)}$.
- 5: Set $k \leftarrow 0$ and $\Delta_P \leftarrow \infty$.
- 6: *Iteration*
- 7: **while** $\Delta_P > \epsilon_P$ **do**
- 8: $k \leftarrow k + 1$.
- 9: Given effective prices $P^{(k-1)}$, solve each worker's expected value function:

$$V_n^{(k)}(o) = \gamma \log \sum_{o'} \exp \left(\frac{P_{o'}^{(k-1)} z_{no'} M_{o'} + s_n(o'|o, \omega_n) + \beta V_n^{(k)}(o')}{\gamma} \right) + \gamma c^e.$$

- 10: Calculate occupation choice probabilities:

$$\pi_n^{(k)}(o'|o) = \frac{\exp \left(\frac{P_{o'}^{(k-1)} z_{no'} M_{o'} + s_n(o'|o, \omega_n) + \beta V_n^{(k)}(o')}{\gamma} \right)}{\sum_j \exp \left(\frac{P_j^{(k-1)} z_{nj} M_j + s_n(j|o, \omega_n) + \beta V_n^{(k)}(j)}{\gamma} \right)}.$$

- 11: Update worker allocation by drawing $o_n^{(k)}$ from $\pi_n^{(k)}(\cdot|o_n^{(k-1)})$.
- 12: Calculate updated occupation outputs $Y_o^{(k)}$ and aggregate output $Y^{(k)}$.
- 13: Update true occupational prices:

$$P_{o,true}^{(k)} = \mu_o^{1/\rho} \left(\frac{Y_o^{(k)}}{Y^{(k)}} \right)^{-1/\rho}.$$

- 14: Update effective occupational prices:

$$P_o^{(k)} \leftarrow \lambda^P P_o^{(k-1)} + (1 - \lambda^P) P_{o,true}^{(k)}.$$

- 15: Calculate price convergence:

$$\Delta_P = \max_o \left| \log P_o^{(k)} - \log P_o^{(k-1)} \right|.$$

- 16: **end while**
-

splitting when two employment episodes overlap. For example, suppose one episode runs from 01/01/2001 to 12/31/2001 and another runs from 09/01/2001 to 09/01/2002. The data provider creates additional episodes for the overlapping period and transfers information from the original episodes to the split records. Episode splitting can mechanically increase measured transitions into and out of the same occupation. To avoid these artificial transitions, I keep an observation only if the episode contains the mid-year date.

I keep observations from two data sources: Employee History (BeH) and Benefit Recipient History (LeH). Benefit Recipient History records individuals receiving unemployment benefits or unemployment assistance. I drop the remaining data sources because they either begin after the start of my sample period or introduce additional complications. Benefit II Recipient History (LHG) starts in 2005, Participants-in-Measures History Files (MTH/XMTH) start in 2000 and 2005, and Jobseeker Histories (ASU/XASU) start in 1997 and 2005. Around half of the observations in Jobseeker Histories come from episode splitting, so I exclude this source from the estimation sample.

Some workers have gaps in their employment histories. Since these gaps make transition probabilities difficult to measure, I keep each worker’s longest continuous history without gaps. If a worker has multiple continuous histories of equal length, I keep the most recent one. I also drop one occupation that accounts for less than 1/1000 of all observations. Finally, I merge two low-observation occupations with related titles: “Occupations in product design, artisan craftwork, fine arts and the making of musical instruments” and “Occupations in the performing arts and entertainment.”

1.A.3.2 Principal Components

Each O*NET SOC occupation is associated with attributes classified under “Knowledge,” “Skills,” “Abilities,” and “Work Activities.” There are 160 attributes in total. Some attributes are closely related, and some concepts, such as “Mathematics,” appear in more than one category. I use principal component analysis to reduce this high-dimensional attribute space. Table 1.A.3.1 reports the largest positive and negative loadings for the first eight principal components. Figure 1.A.3.1 shows where occupations are located in the space of the first two principal components.

Table 1.A.3.1: Most Positive and Negative Loadings for the Principal Components

Principal Component 1			
Most Negative		Most Positive	
Task	Loading	Task	Loading
Manual Dexterity	-0.102	Written Expression	0.114
Extent Flexibility	-0.101	Written Comprehension	0.114
Handling and Moving Objects	-0.100	Writing	0.114
Static Strength	-0.099	Reading Comprehension	0.114
Dynamic Strength	-0.099	Active Learning	0.114

Continued on next page

Table 1.A.3.1 – continued from previous page

Most Negative		Most Positive	
Task	Loading	Task	Loading
Principal Component 2			
Most Negative		Most Positive	
Task	Loading	Task	Loading
Working with the Public	-0.068	Quality Control Analysis	0.148
Customer and Personal Service	-0.053	Mechanical	0.148
Fine Arts	-0.052	Inspecting Equipment, Material	0.149
Service Orientation	-0.052	Operation Monitoring	0.150
Establishing Interpersonal Relat.	-0.042	Physics	0.153
Principal Component 3			
Most Negative		Most Positive	
Task	Loading	Task	Loading
Programming	-0.142	Working with the Public	0.154
Interacting With Computers	-0.130	Resolving Conflicts and Negotiating	0.155
Computers and Electronics	-0.126	Psychology	0.169
Near Vision	-0.110	Assisting and Caring for Others	0.180
Engineering and Technology	-0.102	Therapy and Counseling	0.196
Principal Component 4			
Most Negative		Most Positive	
Task	Loading	Task	Loading
Sales and Marketing	-0.240	Biology	0.139
Management of Material Resources	-0.196	Assisting and Caring for Others	0.151
Management of Financial Resources	-0.195	Identifying Objects, Actions, and Events	0.158
Production and Processing	-0.178	Medicine and Dentistry	0.170
Selling or Influencing Others	-0.178	Documenting/Recording Information	0.242
Principal Component 5			
Most Negative		Most Positive	
Task	Loading	Task	Loading
Fine Arts	-0.329	Economics and Accounting	0.113
History and Archeology	-0.276	Processing Information	0.113
Thinking Creatively	-0.228	Performing Administrative Activities	0.125
Philosophy and Theology	-0.209	Number Facility	0.132
Sociology and Anthropology	-0.185	Determine Compliance	0.173
Principal Component 6			
Most Negative		Most Positive	
Task	Loading	Task	Loading
Geography	-0.321	Monitor Processes, Materials	0.113
Transportation	-0.302	Finger Dexterity	0.116
Telecommunications	-0.238	Arm-Hand Steadiness	0.118
Law and Government	-0.205	Instructing	0.120
Spatial Orientation	-0.198	Training and Teaching Others	0.144
Principal Component 7			
Most Negative		Most Positive	
Task	Loading	Task	Loading

Continued on next page

Table 1.A.3.1 – continued from previous page

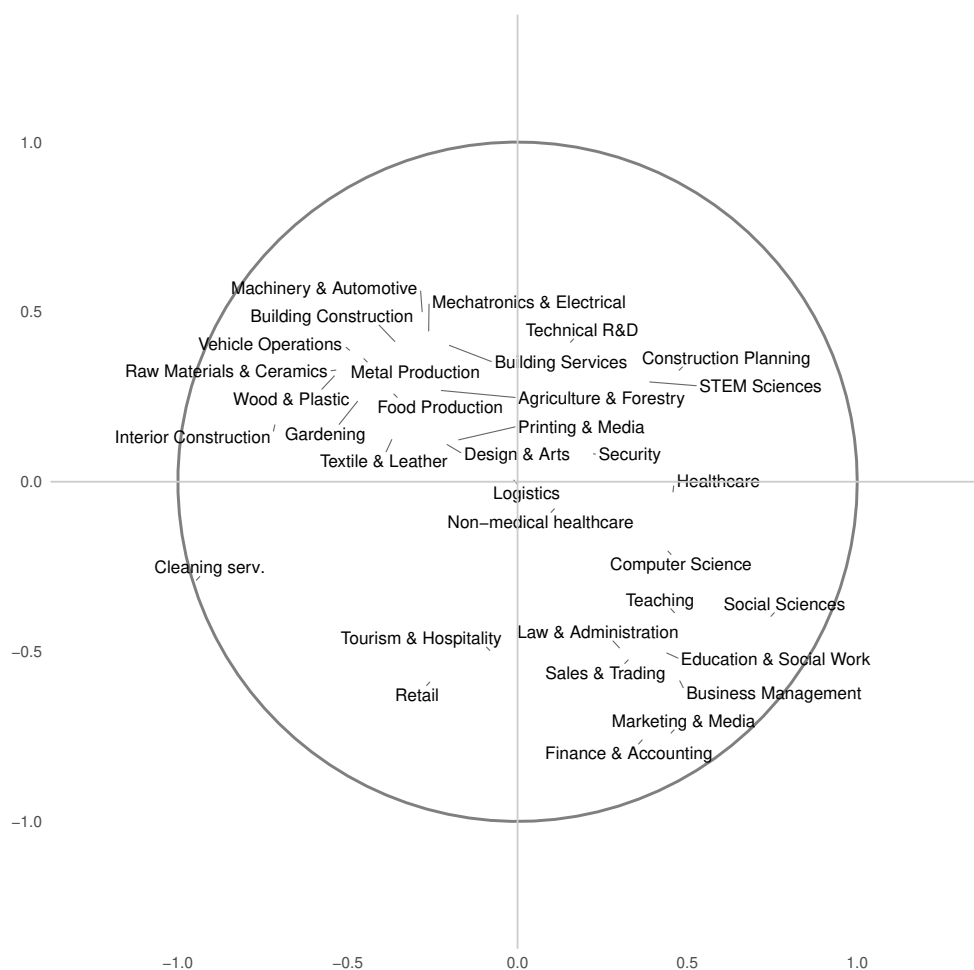
Most Negative		Most Positive	
Task	Loading	Task	Loading
Task	Loading	Task	Loading
Food Production	-0.218	Time Sharing	0.185
Biology	-0.162	Finger Dexterity	0.190
Estimating the Characteristics of Info.	-0.131	Telecommunications	0.234
Dynamic Flexibility	-0.114	Clerical	0.238
Geography	-0.100	Customer and Personal Service	0.259

Principal Component 8

Most Negative		Most Positive	
Task	Loading	Task	Loading
Chemistry	-0.243	Peripheral Vision	0.117
Biology	-0.241	Speech Clarity	0.118
Economics and Accounting	-0.240	Installation	0.141
Customer and Personal Service	-0.219	Sound Localization	0.148
Sales and Marketing	-0.190	Selective Attention	0.152

Note: I use the first 8 principal components in the estimation of the comparative advantage parameters, which explain more than 80 percent of the total variance in the entire space.

Figure 1.A.3.1: Occupations in the First Two Principal Component Dimensions



Note: x -axis is the first principal component and y -axis is the second principal component. The principal components are scaled so that all occupations would be in the interior of the unit circle.

1.A.3.3 Crosswalks

There is no direct crosswalk between O*NET SOC 2010 and KldB 2010. I therefore map occupations from SOC 2010 to ISCO-08 and then from ISCO-08 to KldB 2010.

I construct crosswalks for automation scores and principal components. The procedure is similar in both cases. The SOC 2010 to ISCO-08 crosswalk maps 6-digit SOC occupations to 4-digit ISCO-08 occupations. For automation scores, I use the task statements associated with each 6-digit SOC occupation when they are available in O*NET. If task statements are not available, I use task statements from child occupations in the SOC hierarchy. If child occupations are also unavailable, I use task statements from sibling occupations.

For principal components, I use the same hierarchy. Instead of combining task statements, I take simple averages of the principal component values. The average is taken over child occupations when available and over sibling occupations otherwise.

The final data contain 34 2-digit KldB 2010 occupations, excluding army occupations and two occupations removed because of low observation counts. There are roughly 1,000 6-digit SOC occupations. Since each KldB occupation is matched to more than 20 SOC occupations on average, cases where I rely on child or sibling occupations are unlikely to drive the aggregate occupation measures.

1.A.4 LLM Prompt

Following is the initial prompt given to Gemini 2.5 Flash model.

Consider the most powerful Google Gemini large language model (LLM). This model can complete many tasks that can be formulated as having text/audio/video input and text/audio/video output. This model have access to up-to-date facts from internet or any information or database that is relevant for the task.

You are a helpful assistant who wants to label the given tasks according to the rubric below. Equivalent quality means someone reviewing the work would not be able to tell whether a human completed it on their own or with assistance from the LLM. If you aren't sure how to judge the amount of time a task takes, consider whether the tools described exposed the majority of subtasks associated with the task.

Exposure rubric:

E1 - Direct exposure: Label tasks E1 if direct access to the LLM through an interface alone can reduce the time it takes to complete the task with equivalent quality by at least half. This includes tasks that can be reduced to: - Writing and transforming text and code according to complex instructions, - Providing edits to existing text or code following specifications, - Writing code that can help perform a task that used to be done by hand, - Translating text between languages, - Summarizing documents, - Providing feedback on documents, - Answering questions about a document, - Generating questions a user might want to ask about a document, - Writing questions for an interview or assessment, - Prepare training materials based on general knowledge, - Inform anyone of any information via written or spoken media.

E0 - No exposure: Label tasks E0 if none of the above clearly decrease the time it takes for an experienced worker to complete the task with high quality by at least half. Some examples: - If a task requires a high degree of human interaction (for example, in-person demonstrations) then it should be classified as E0. - If a task requires precise measurements then it should be classified as E0. - If a task requires any use of a hand or walking then it should be classified as E0. - LLM cannot make any decisions that might impact human livelihood (e.g. hiring, grading, etc.). If any part of the task involves collecting inputs to make a final decision (as opposed to analyzing data to inform a decision or make a recommendation) then it should be classified as E0. The LLM can make recommendations. The LLM cannot do anything that legally requires a human to perform the task. - If there is existing technology not powered

by an LLM that is commonly used and can complete the task then you should mark the task E0 if using an LLM or LLM-powered tool will not further reduce the time to complete the task. When in doubt, you should default to E0.

Read the following tasks that belong to an occupation. Then do the following: 1. Reason very briefly step by step to decide which of the labels (E0 or E1) from the exposure rubric you were given applies to the task's exposure to LLM. Report. 2. Give an explanation for which label applies and report the label that you think fits best. Report only the label that you determined for the task. 3: 3. Given the amount of speculation required in step 1, describe your certainty about the estimate – either high, moderate, or low.

Direct or coordinate an organization's financial or budget activities to fund operations, maximize investments, or increase efficiency.

LLM Response: *The task 'Direct or coordinate an organization's financial or budget activities to fund operations, maximize investments, or increase efficiency' involves high-level strategic decision-making, leadership, and ultimate responsibility for financial outcomes. These decisions directly impact the organization's stability, employee livelihoods, and overall business operations. The rubric explicitly states: 'LLM cannot make any decisions that might impact human livelihood (e.g. hiring, grading, etc.). If any part of the task involves collecting inputs to make a final decision (as opposed to analyzing data to inform a decision or make a recommendation) then it should be classified as E0.' While an LLM can analyze financial data, generate reports, draft proposals, or make recommendations (which are E1 activities), it cannot 'direct or coordinate' by making the final, responsible decisions or taking on the leadership role required for such a task. Furthermore, this task requires a high degree of human interaction, negotiation, and strategic judgment that an LLM cannot replicate with equivalent quality. Therefore, the core of the task is not exposed to LLM capabilities in a way that would reduce completion time by half with equivalent quality. E0 High*

...

...

...

I use the “chat” feature in the API, which allows me to respond to every output by the next task description. This allows me to save fees and time by not having to input the initial prompt every time.